# Result Compendium

$Z \to \ell\ell\gamma$ — Unconverted $\gamma$

$\eta$-only binning, tight isolation

Cell-Energy Reweighting Pipeline

May 3, 2026

## Contents


**1 Chi-Squared Summary** **2**

**2 Integrated Shower Shape Comparisons** **3**

**3 Per-$|\eta|$ Shower Shape Comparisons** **4**

**4 Shower Shape Comparisons vs. ATLAS Fudge Factors** **9**

**5 Fudge Factor Comparison (Integrated)** **14**

**6 Fudge Factor Comparison (Per $|\eta|$ Bin)** **15**

**7 Computed vs. Stored Validation (Integrated)** **20**

**8 Computed vs. Stored (Per $|\eta|$ Bin)** **21**

**9 Cell Energy Fraction Profiles** **26**

**10 Cell-Level Corrections** **39**

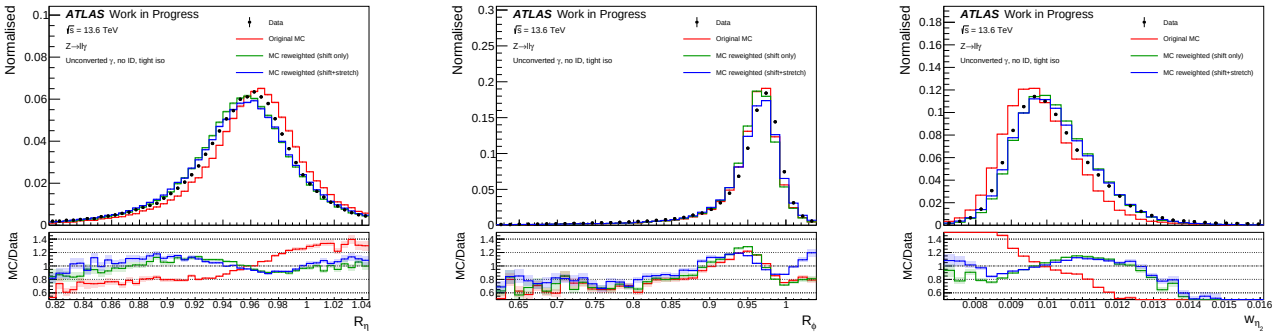
# 1 Chi-Squared Summary

Table 1: Per-$|\eta|$-bin $\chi^2/n_{\mathrm{df}}$ for $Z \to \ell\ell\gamma$, Unconverted scenario. Columns show $\chi^2/n_{\mathrm{df}}$ between each MC variant and data (cell-computed).

| $\lvert\eta\rvert$ bin | $R_\eta$ | | | | $R_\phi$ | | | | $w_{\eta 2}$ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (range) | MC | M1 | M2 | Fud | MC | M1 | M2 | Fud | MC | M1 | M2 | Fud |
| [0.00, 0.20) | 0.0003 | 0.0003 | 0.0002 | 0.0002 | 0.0004 | 0.0005 | 0.0003 | 0.0002 | 0.0008 | 0.0003 | 0.0002 | 0.0003 |
| [0.20, 0.40) | 0.0002 | 0.0001 | 0.0001 | 0.0001 | 0.0003 | 0.0004 | 0.0003 | 0.0002 | 0.0016 | 0.0003 | 0.0003 | 0.0003 |
| [0.40, 0.60) | 0.0003 | 0.0002 | 0.0002 | 0.0001 | 0.0005 | 0.0004 | 0.0004 | 0.0002 | 0.0014 | 0.0004 | 0.0003 | 0.0003 |
| [0.60, 0.80) | 0.0003 | 0.0002 | 0.0002 | 0.0002 | 0.0003 | 0.0006 | 0.0003 | 0.0002 | 0.0015 | 0.0003 | 0.0002 | 0.0003 |
| [0.80, 1.00) | 0.0005 | 0.0002 | 0.0003 | 0.0002 | 0.0007 | 0.0005 | 0.0003 | 0.0003 | 0.0006 | 0.0004 | 0.0002 | 0.0004 |
| [1.00, 1.20) | 0.0003 | 0.0002 | 0.0002 | 0.0002 | 0.0010 | 0.0011 | 0.0005 | 0.0003 | 0.0005 | 0.0005 | 0.0003 | 0.0003 |
| [1.20, 1.30) | 0.0006 | 0.0031 | 0.0031 | 0.0004 | 0.0008 | 0.0008 | 0.0004 | 0.0005 | 0.0010 | 0.0047 | 0.0038 | 0.0004 |
| [1.30, 1.37) | 0.0028 | 0.0013 | 0.0017 | 0.0021 | 0.0014 | 0.0018 | 0.0013 | 0.0016 | 0.0024 | 0.0022 | 0.0017 | 0.0012 |
| [1.52, 1.60) | 0.0016 | 0.0032 | 0.0019 | 0.0009 | 0.0014 | 0.0038 | 0.0012 | 0.0013 | 0.0068 | 0.0015 | 0.0011 | 0.0010 |
| [1.60, 1.80) | 0.0014 | 0.0006 | 0.0005 | 0.0003 | 0.0004 | 0.0013 | 0.0006 | 0.0005 | 0.0083 | 0.0004 | 0.0005 | 0.0004 |
| [1.80, 2.00) | 0.0022 | 0.0006 | 0.0012 | 0.0004 | 0.0006 | 0.0013 | 0.0011 | 0.0005 | 0.0164 | 0.0014 | 0.0007 | 0.0005 |
| [2.00, 2.20) | 0.0034 | 0.0004 | 0.0005 | 0.0004 | 0.0005 | 0.0010 | 0.0008 | 0.0004 | 0.0252 | 0.0005 | 0.0004 | 0.0004 |
| [2.20, 2.40) | 0.0056 | 0.0008 | 0.0009 | 0.0006 | 0.0012 | 0.0018 | 0.0009 | 0.0009 | 0.0319 | 0.0010 | 0.0006 | 0.0005 |
| Average | 0.0015 | 0.0009 | 0.0008 | 0.0005 | 0.0007 | 0.0012 | 0.0006 | 0.0005 | 0.0076 | 0.0011 | 0.0008 | 0.0005 |

## 2 Integrated Shower Shape Comparisons

Data versus uncorrected MC, M1, and M2 corrections (all $|\eta|$ bins combined).



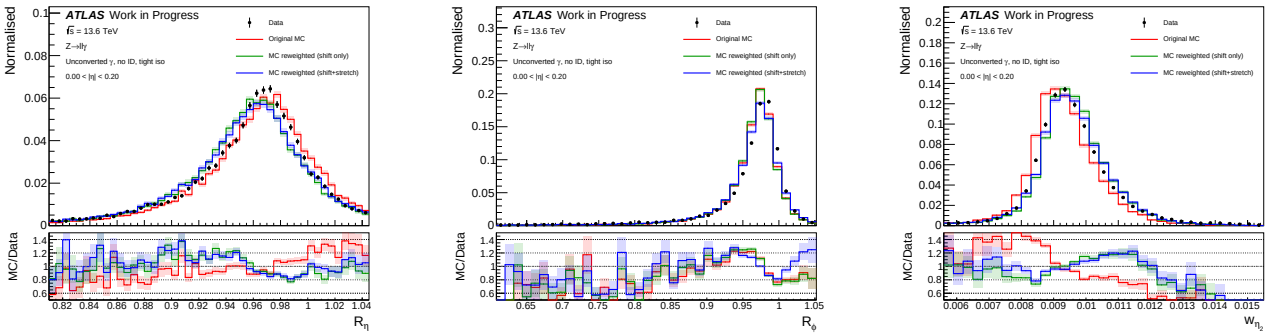

(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 1: Integrated shower shape comparisons: Data vs uncorrected MC, M1 and M2.

# 3 Per-$|\eta|$ Shower Shape Comparisons

Data versus uncorrected MC, M1, and M2 corrections for each $|\eta|$ bin.

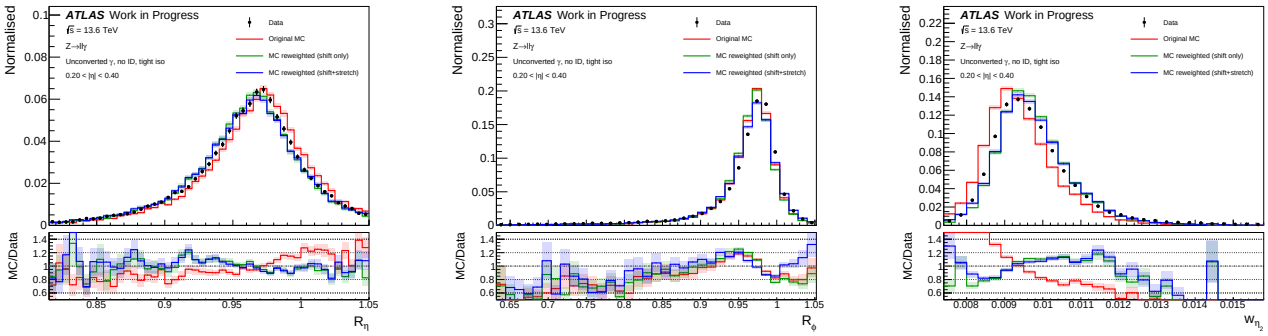
$|\eta| \in [0.00, 0.20)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 2: Shower shape comparisons, $|\eta| \in [0.00, 0.20)$.

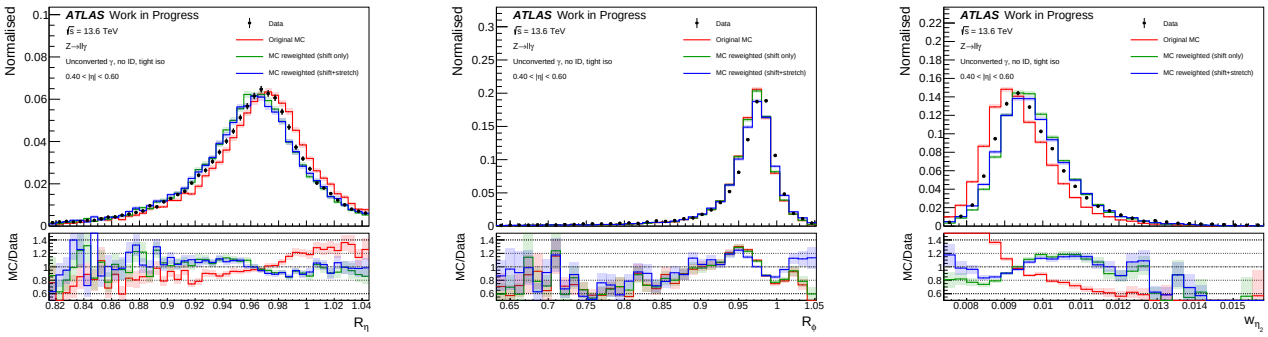
$|\eta| \in [0.20, 0.40)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 3: Shower shape comparisons, $|\eta| \in [0.20, 0.40)$.

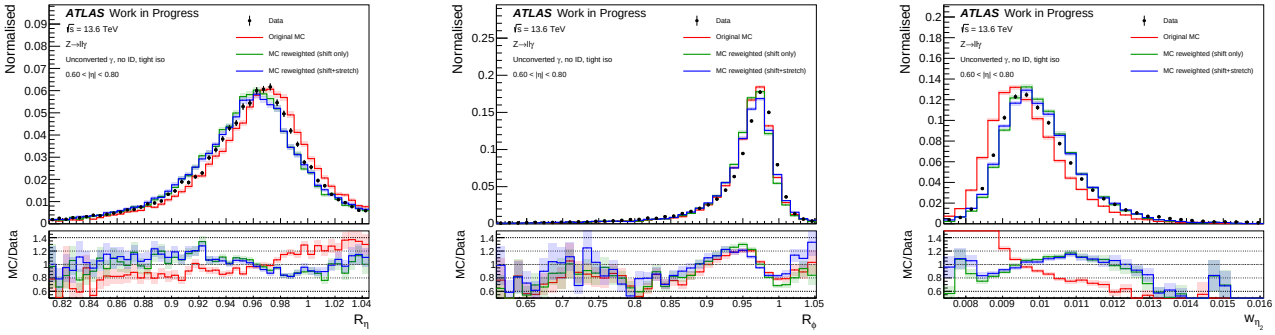
$|\eta| \in [0.40, 0.60)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 4: Shower shape comparisons, $|\eta| \in [0.40, 0.60)$.

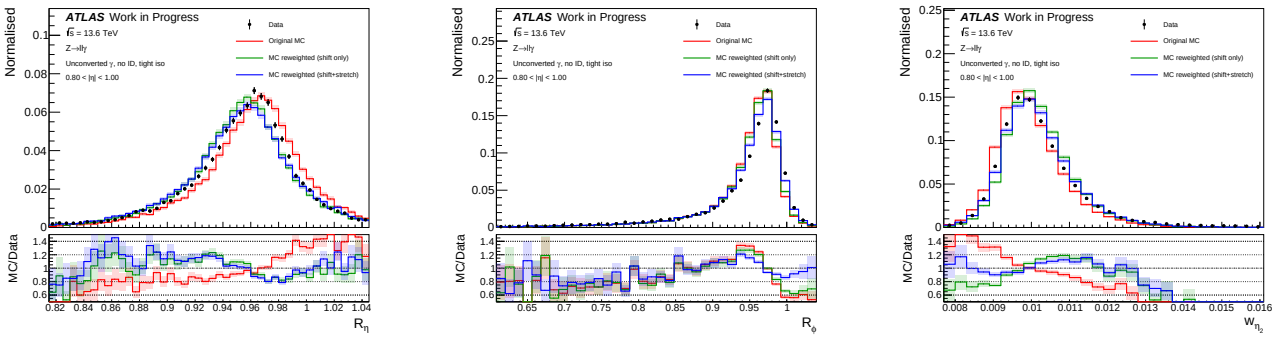
$|\eta| \in [0.60, 0.80)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 5: Shower shape comparisons, $|\eta| \in [0.60, 0.80)$.

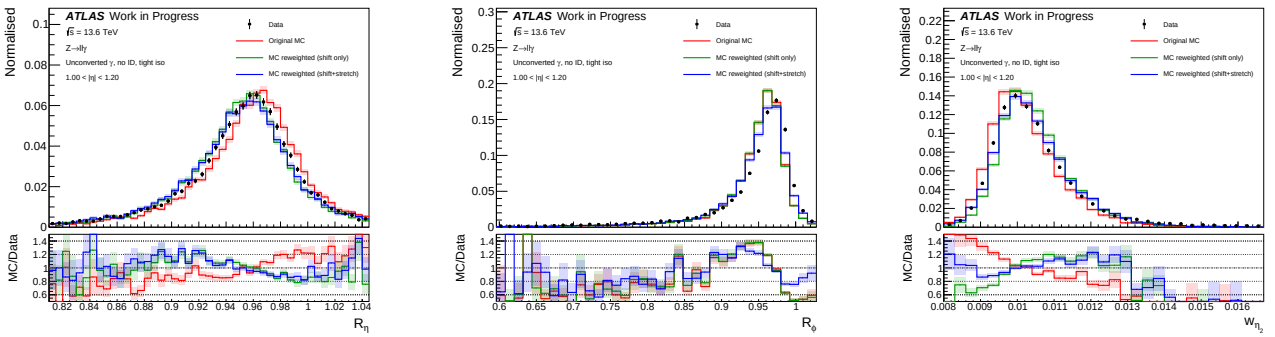
$|\eta| \in [0.80, 1.00)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 6: Shower shape comparisons, $|\eta| \in [0.80, 1.00)$.

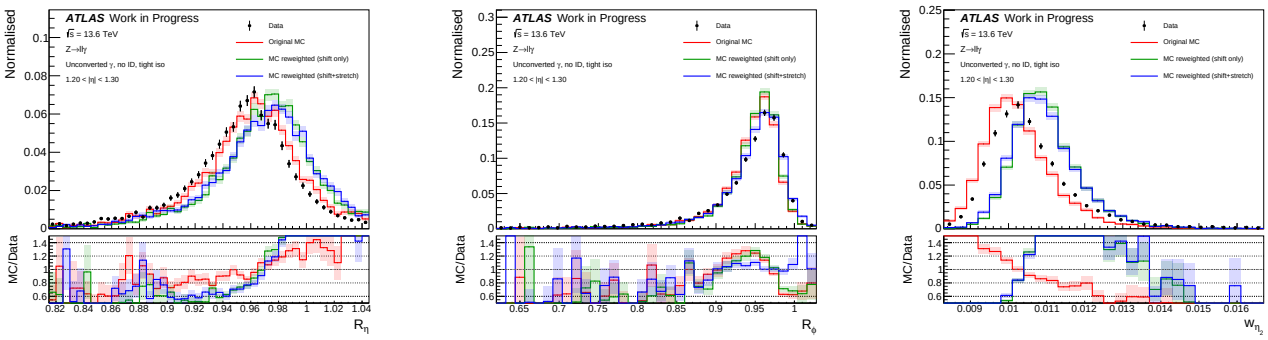
$|\eta| \in [1.00, 1.20)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 7: Shower shape comparisons, $|\eta| \in [1.00, 1.20)$.

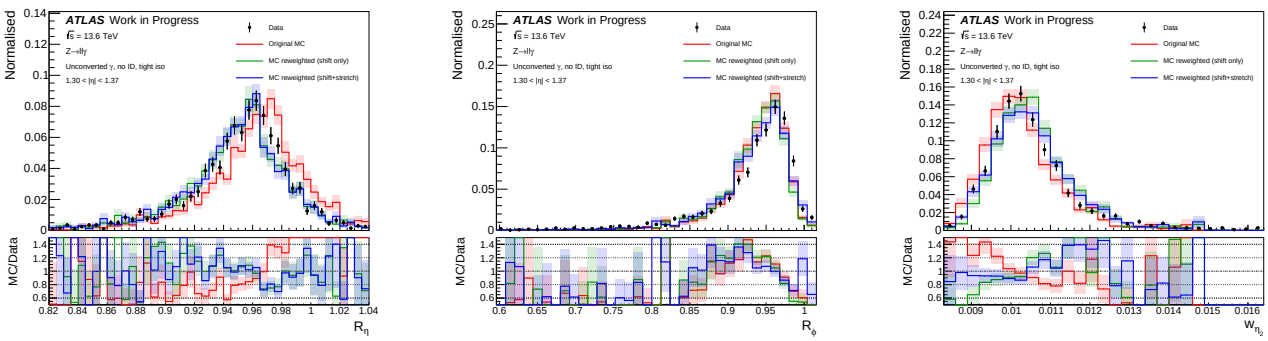
$|\eta| \in [1.20, 1.30)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 8: Shower shape comparisons, $|\eta| \in [1.20, 1.30)$.

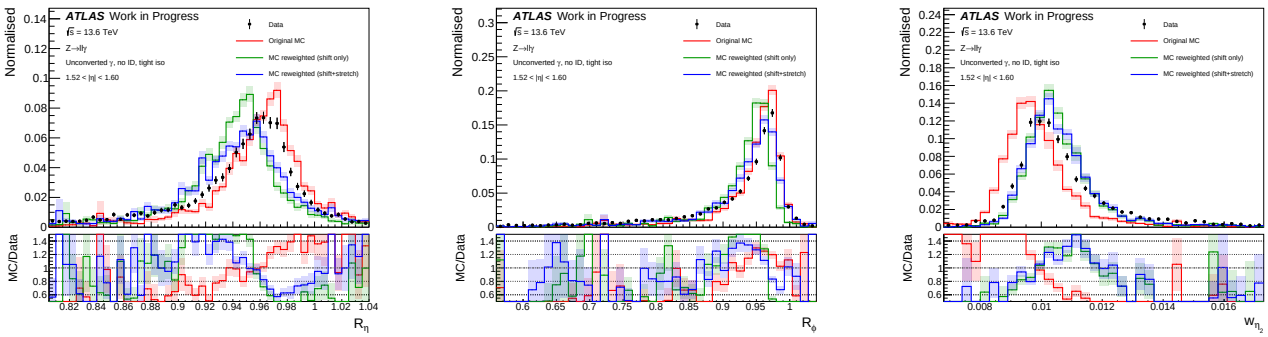
$|\eta| \in [1.30, 1.37)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 9: Shower shape comparisons, $|\eta| \in [1.30, 1.37)$.

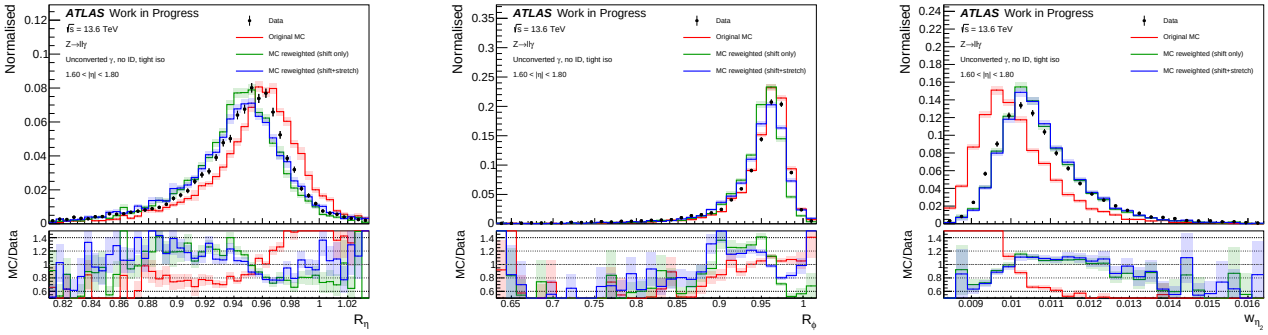
$|\eta| \in [1.52, 1.60)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 10: Shower shape comparisons, $|\eta| \in [1.52, 1.60)$.

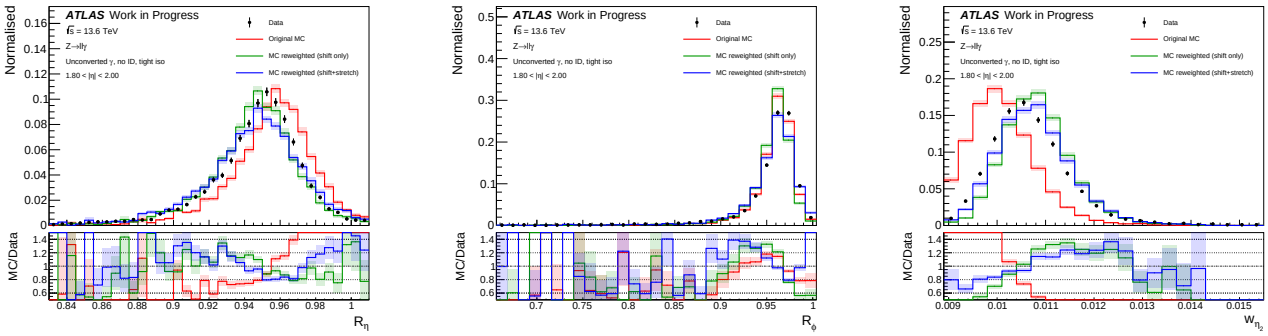
$|\eta| \in [1.60, 1.80)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 11: Shower shape comparisons, $|\eta| \in [1.60, 1.80)$.

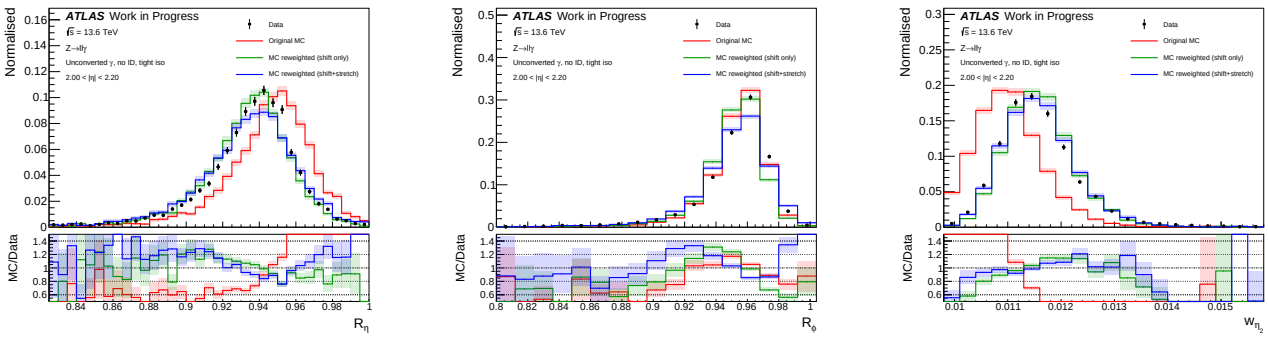
$|\eta| \in [1.80, 2.00)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 12: Shower shape comparisons, $|\eta| \in [1.80, 2.00)$.

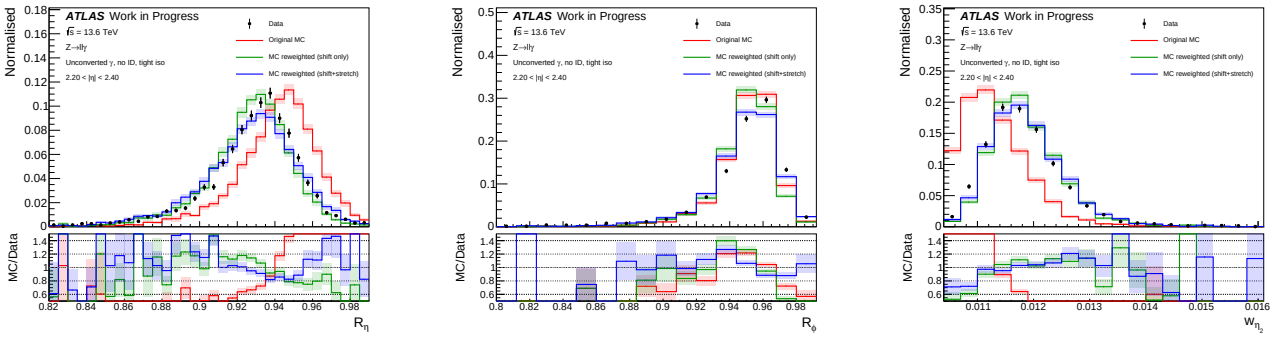
$|\eta| \in [2.00, 2.20)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 13: Shower shape comparisons, $|\eta| \in [2.00, 2.20)$.

$|\eta| \in [2.20, 2.40)$



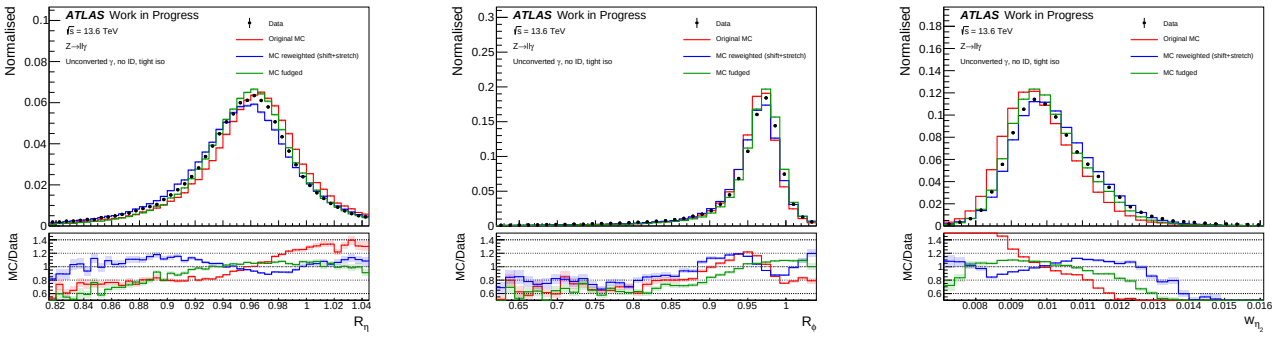

(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 14: Shower shape comparisons, $|\eta| \in [2.20, 2.40)$.

# 4 Shower Shape Comparisons vs. ATLAS Fudge Factors

Data (cell-computed) versus original MC, M2 cell reweighting, and ATLAS fudge factors (stored-branch fudged MC). Provides a direct comparison of the cell-reweighting correction against the existing ATLAS shower-shape corrections.

## Integrated (all $|\eta|$ bins)

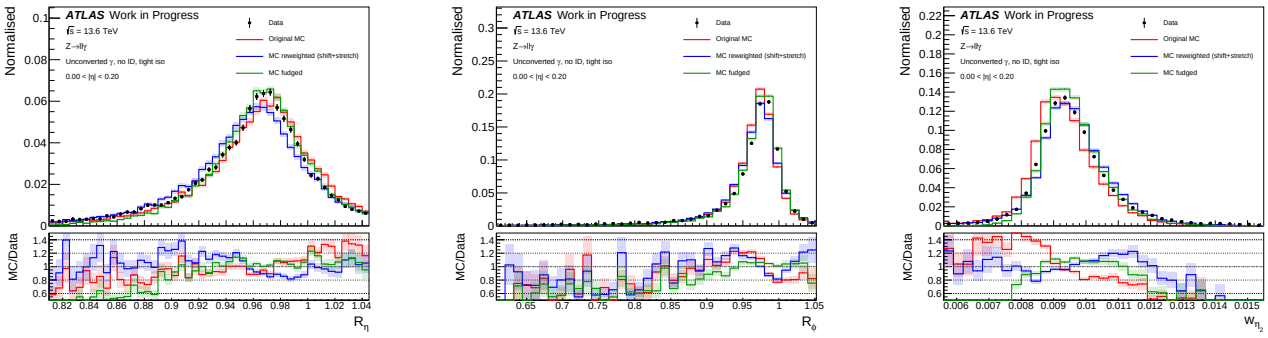



(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 15: Integrated: Data vs. original MC, M2 reweighting, and ATLAS fudged MC.

## Per $|\eta|$ bin

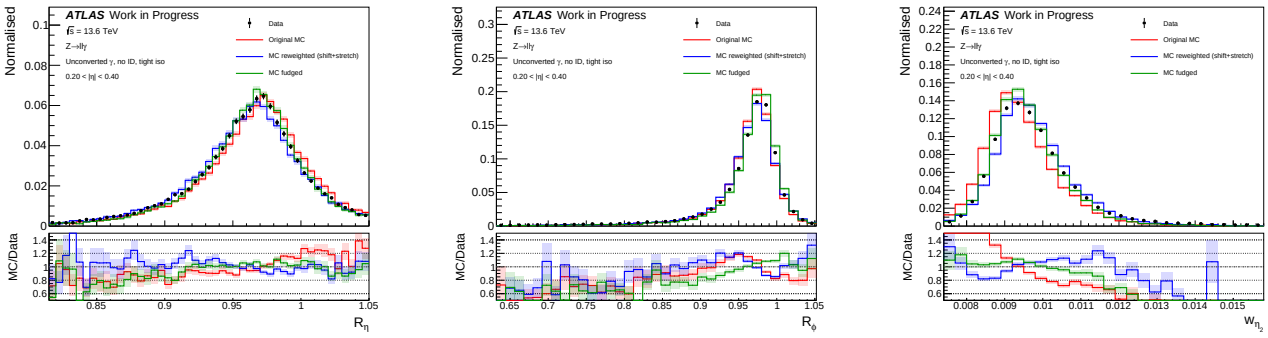
$|\eta| \in [0.00, 0.20)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 16: Shower shapes vs. fudge factors, $|\eta| \in [0.00, 0.20)$.

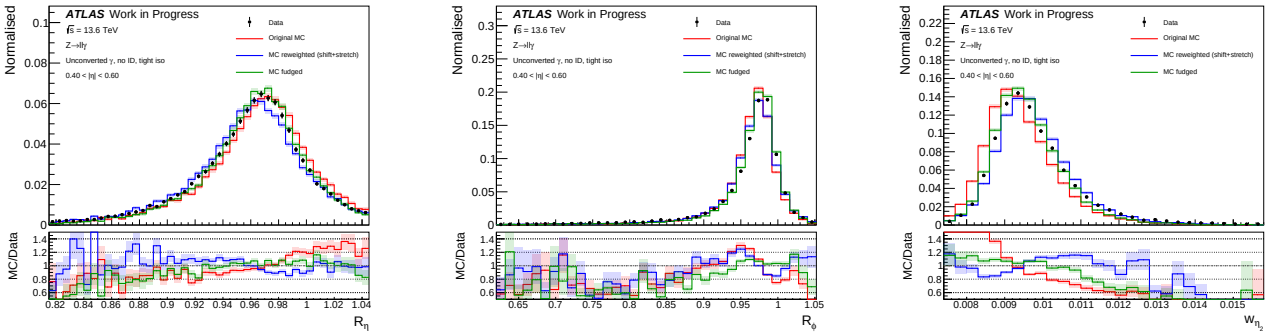
$|\eta| \in [0.20, 0.40)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 17: Shower shapes vs. fudge factors, $|\eta| \in [0.20, 0.40)$.

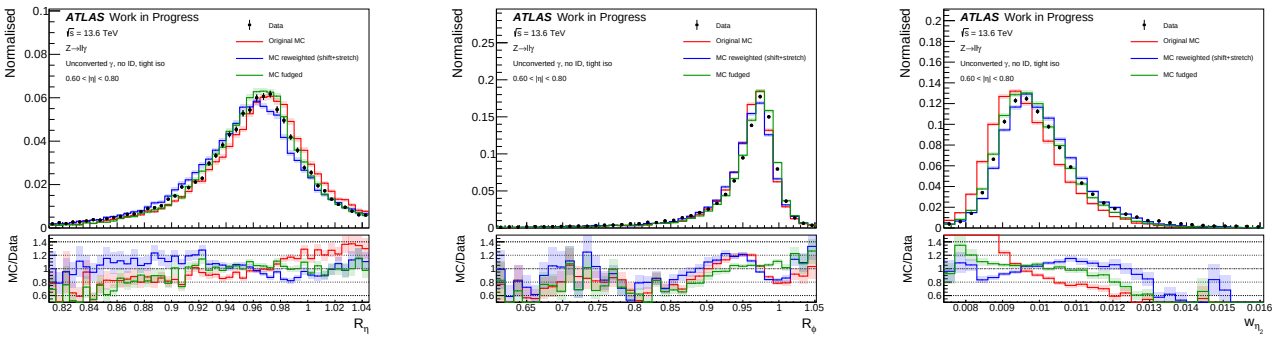
$|\eta| \in [0.40, 0.60)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 18: Shower shapes vs. fudge factors, $|\eta| \in [0.40, 0.60)$.

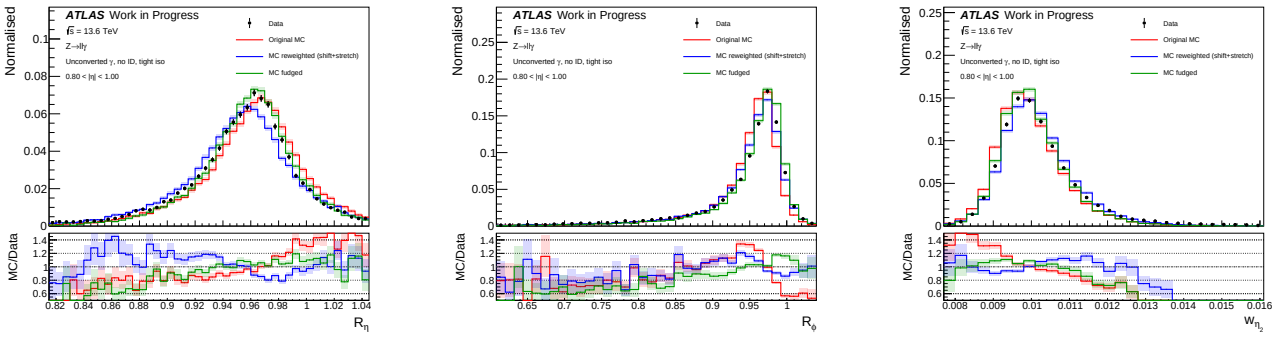
$|\eta| \in [0.60, 0.80)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 19: Shower shapes vs. fudge factors, $|\eta| \in [0.60, 0.80)$.

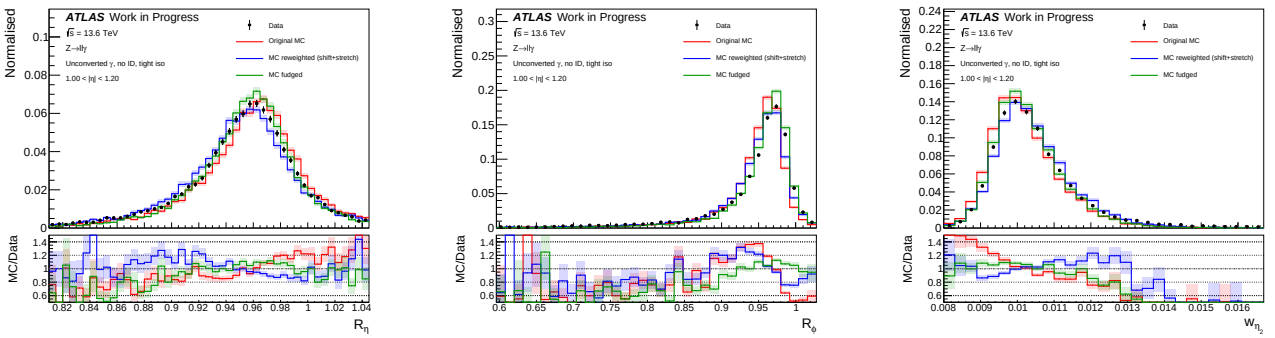
$|\eta| \in [0.80, 1.00)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 20: Shower shapes vs. fudge factors, $|\eta| \in [0.80, 1.00)$.

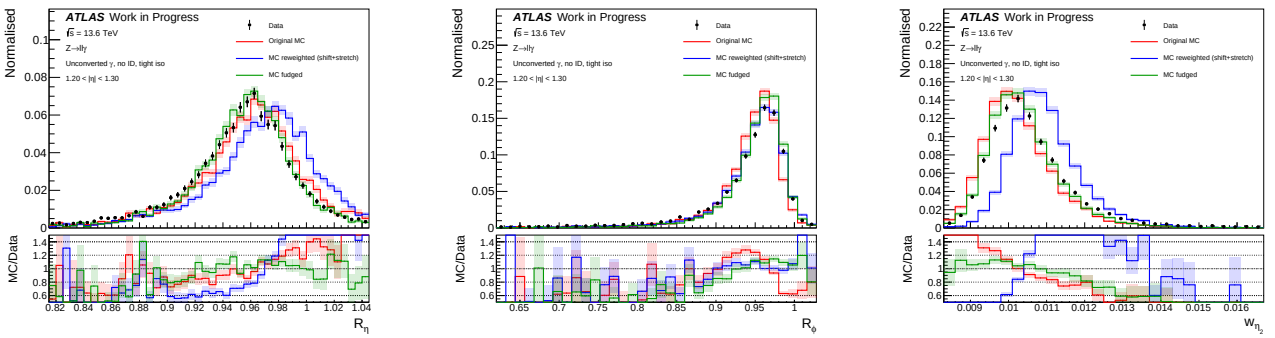
$|\eta| \in [1.00, 1.20)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 21: Shower shapes vs. fudge factors, $|\eta| \in [1.00, 1.20)$.

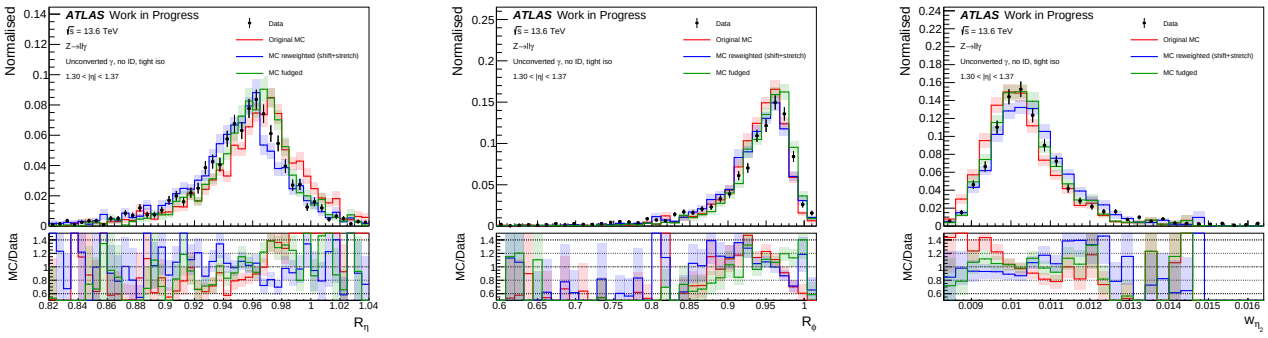
$|\eta| \in [1.20, 1.30)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 22: Shower shapes vs. fudge factors, $|\eta| \in [1.20, 1.30)$.

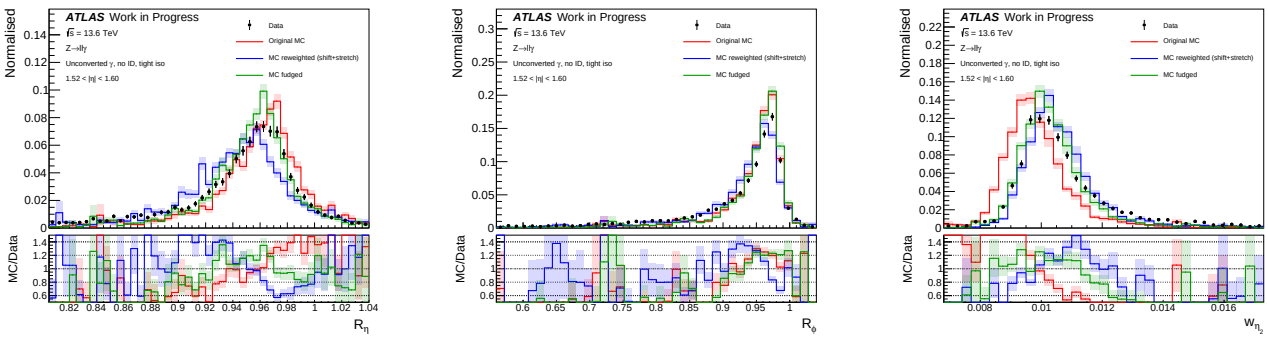
$|\eta| \in [1.30, 1.37)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 23: Shower shapes vs. fudge factors, $|\eta| \in [1.30, 1.37)$.

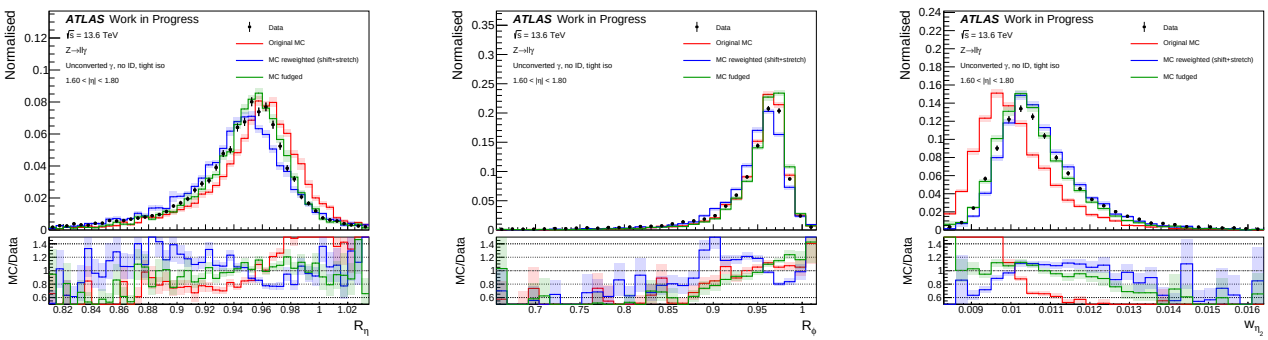
$|\eta| \in [1.52, 1.60)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 24: Shower shapes vs. fudge factors, $|\eta| \in [1.52, 1.60)$.

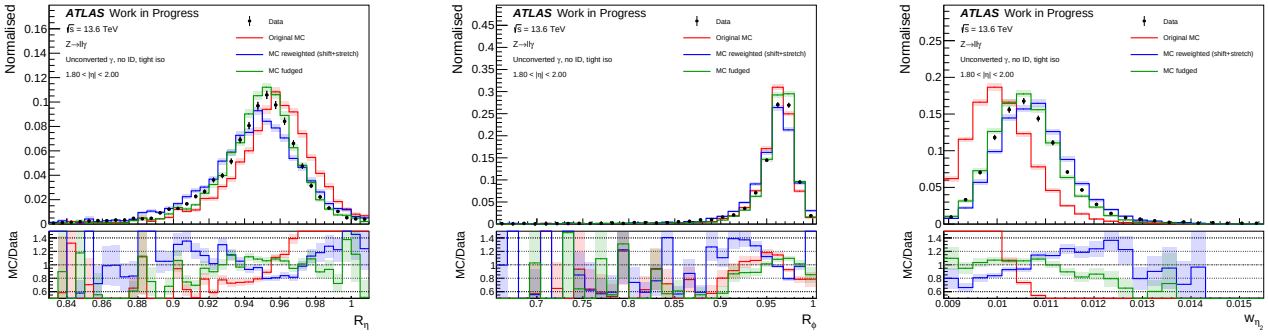
$|\eta| \in [1.60, 1.80)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 25: Shower shapes vs. fudge factors, $|\eta| \in [1.60, 1.80)$.

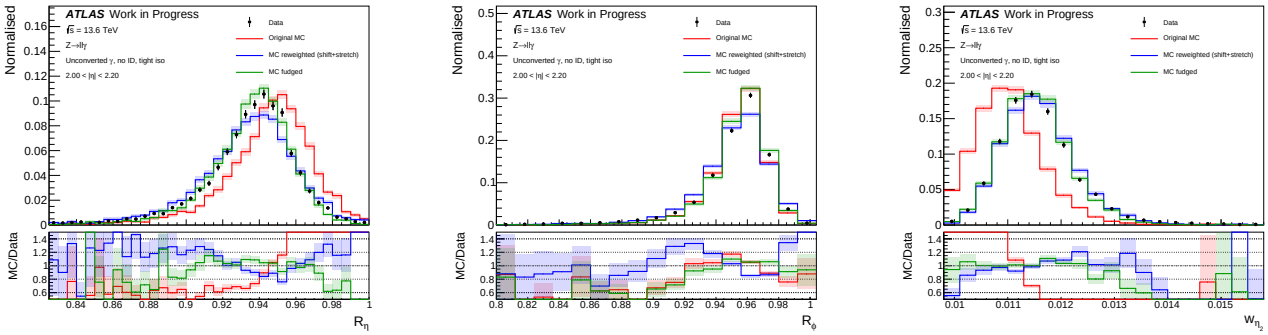
$|\eta| \in [1.80, 2.00)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 26: Shower shapes vs. fudge factors, $|\eta| \in [1.80, 2.00)$.

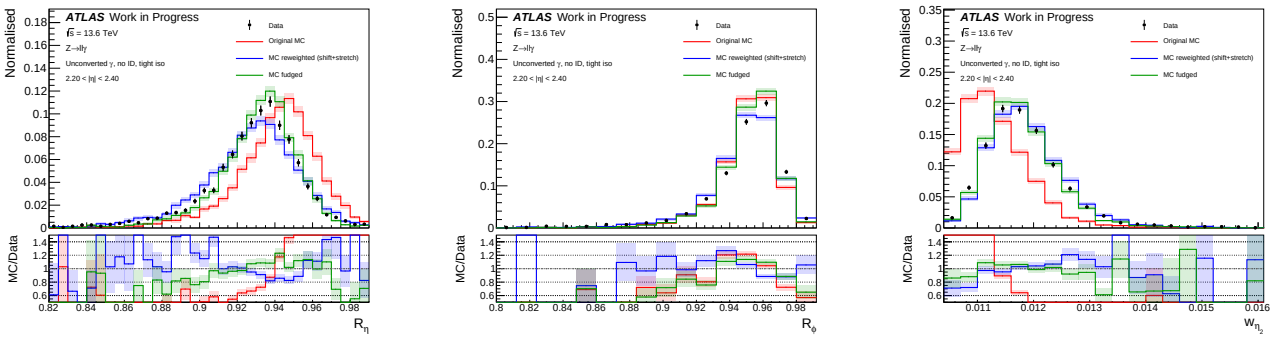
$|\eta| \in [2.00, 2.20)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 27: Shower shapes vs. fudge factors, $|\eta| \in [2.00, 2.20)$.

$|\eta| \in [2.20, 2.40)$

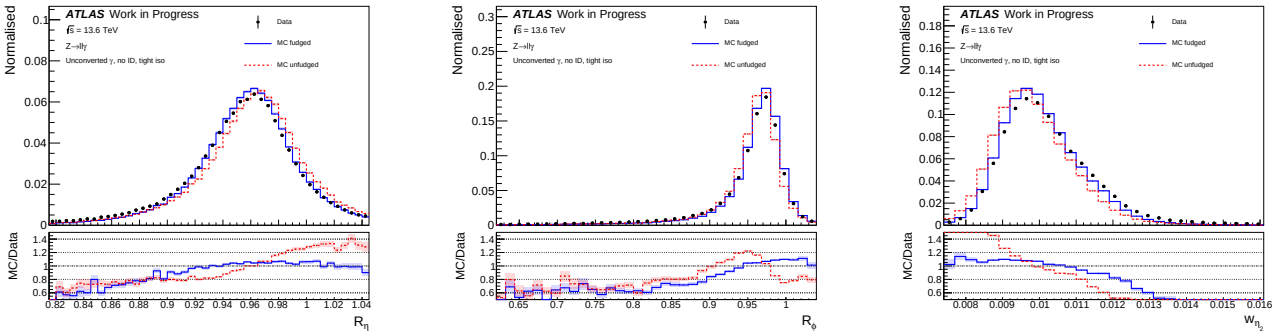



(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 28: Shower shapes vs. fudge factors, $|\eta| \in [2.20, 2.40)$.

# 5 Fudge Factor Comparison (Integrated)

Stored (fudged) branch values versus unfudged MC and data.



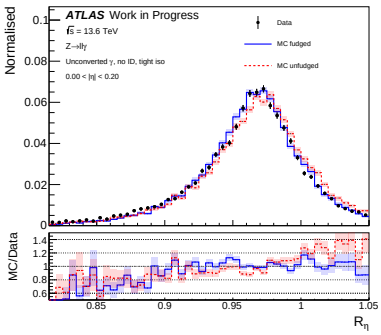

(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 29: Fudge factor comparison: Data (stored) vs MC (fudged) vs MC (unfudged).

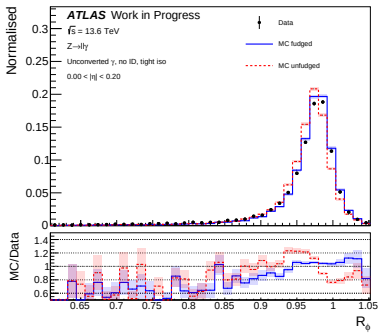
# 6 Fudge Factor Comparison (Per $|\eta|$ Bin)

Fudge factor comparisons for each $|\eta|$ bin.

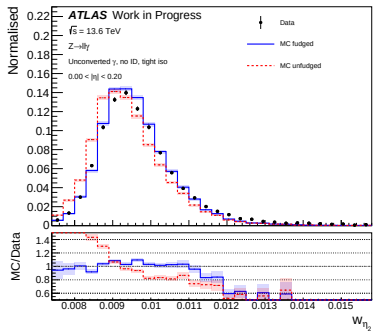
$|\eta| \in [0.00, 0.20)$




(a) $R_\eta$



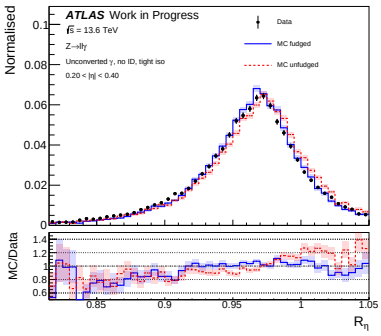

(b) $R_\phi$



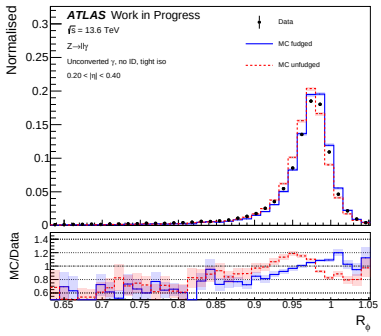

(c) $w_{\eta 2}$

Figure 30: Fudge factor comparison, $|\eta| \in [0.00, 0.20)$.

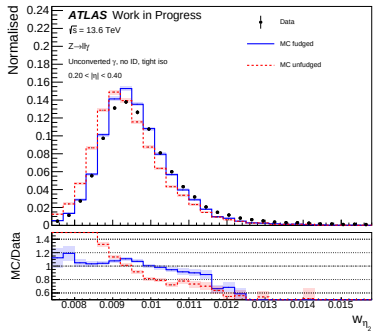
$|\eta| \in [0.20, 0.40)$




(a) $R_\eta$



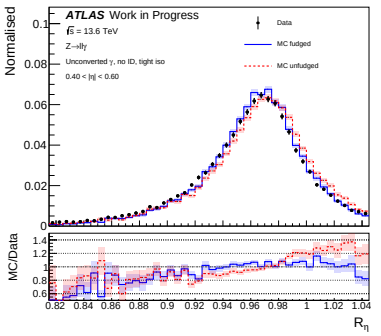

(b) $R_\phi$



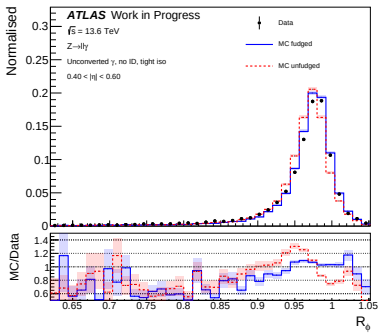

(c) $w_{\eta 2}$

Figure 31: Fudge factor comparison, $|\eta| \in [0.20, 0.40)$.

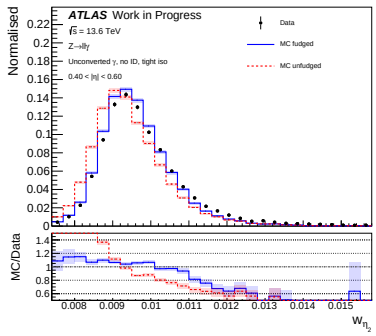
$|\eta| \in [0.40, 0.60)$




(a) $R_\eta$



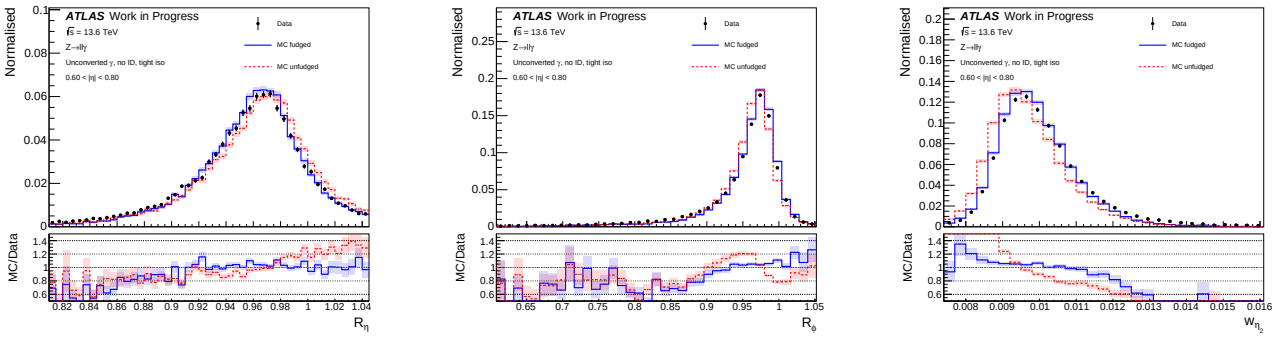

(b) $R_\phi$




(c) $w_{\eta 2}$

Figure 32: Fudge factor comparison, $|\eta| \in [0.40, 0.60)$.

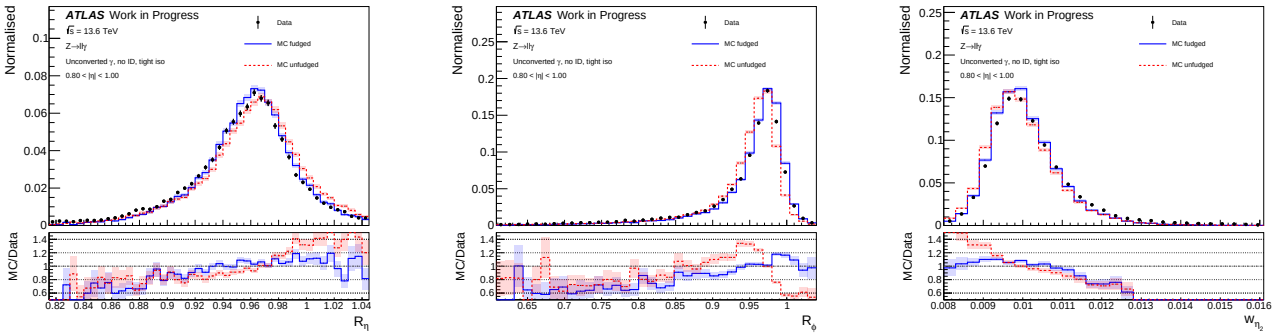
$|\eta| \in [0.60, 0.80)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 33: Fudge factor comparison, $|\eta| \in [0.60, 0.80)$.

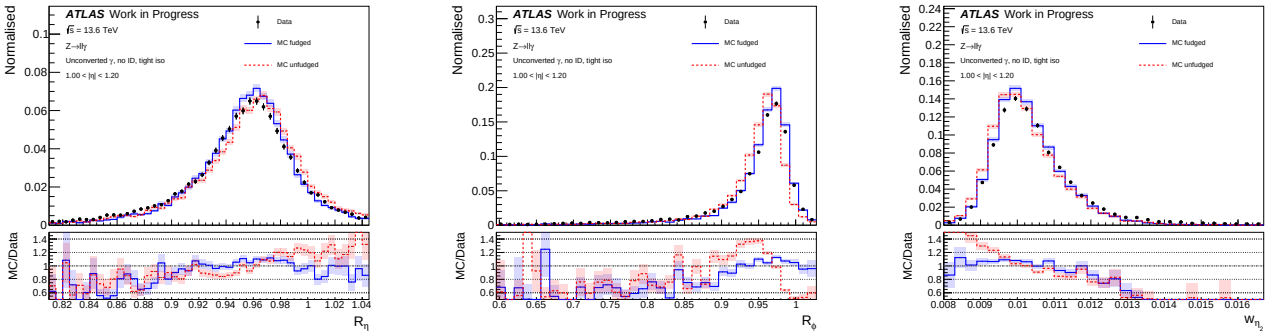
$|\eta| \in [0.80, 1.00)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 34: Fudge factor comparison, $|\eta| \in [0.80, 1.00)$.

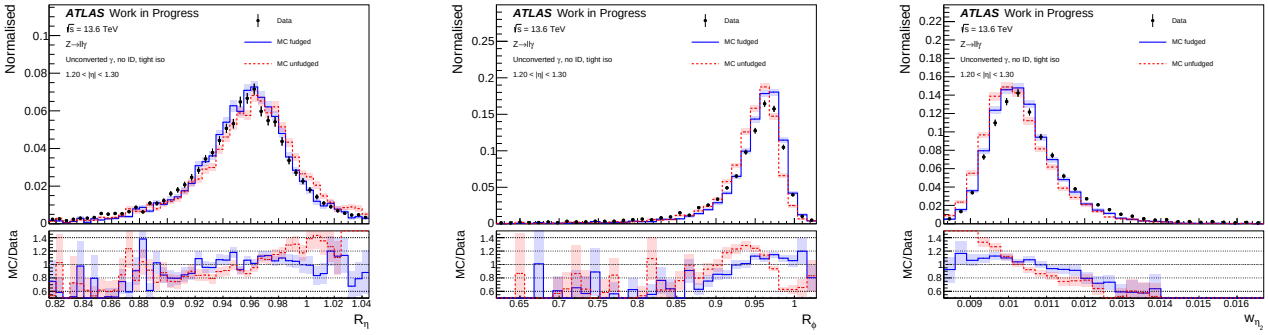
$|\eta| \in [1.00, 1.20)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 35: Fudge factor comparison, $|\eta| \in [1.00, 1.20)$.

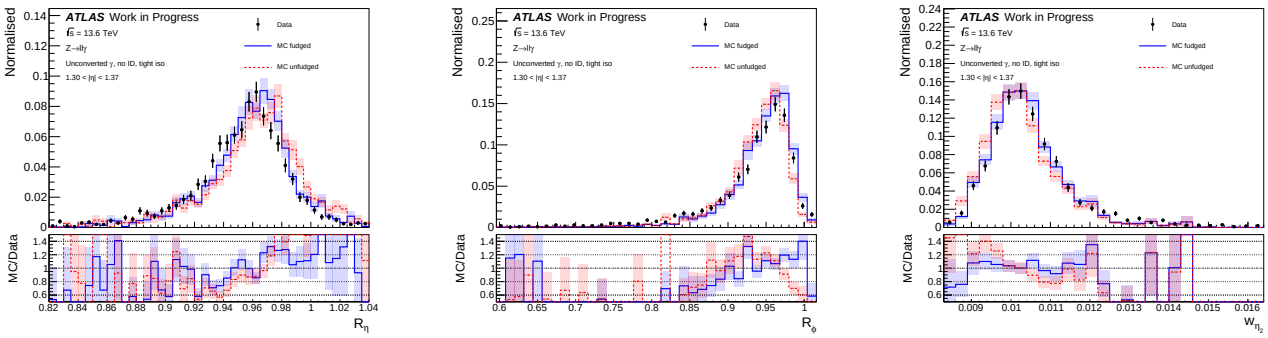
$|\eta| \in [1.20, 1.30)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 36: Fudge factor comparison, $|\eta| \in [1.20, 1.30)$.

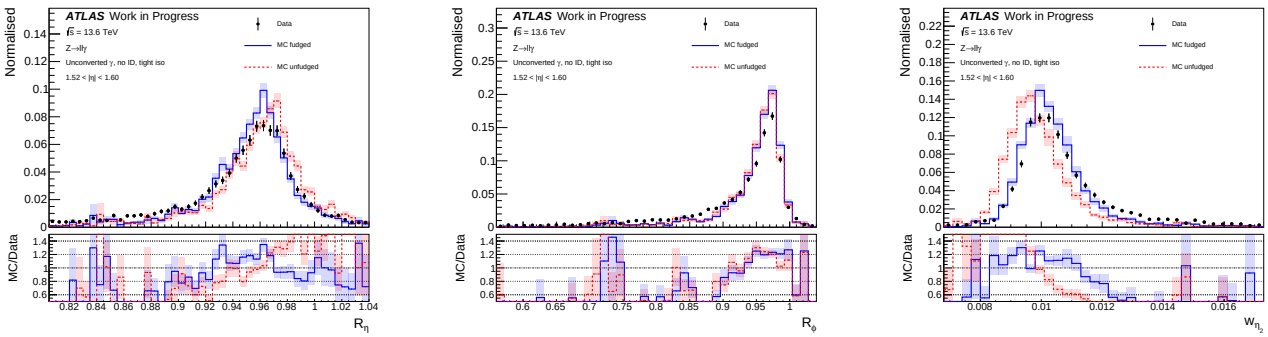
$|\eta| \in [1.30, 1.37)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 37: Fudge factor comparison, $|\eta| \in [1.30, 1.37)$.

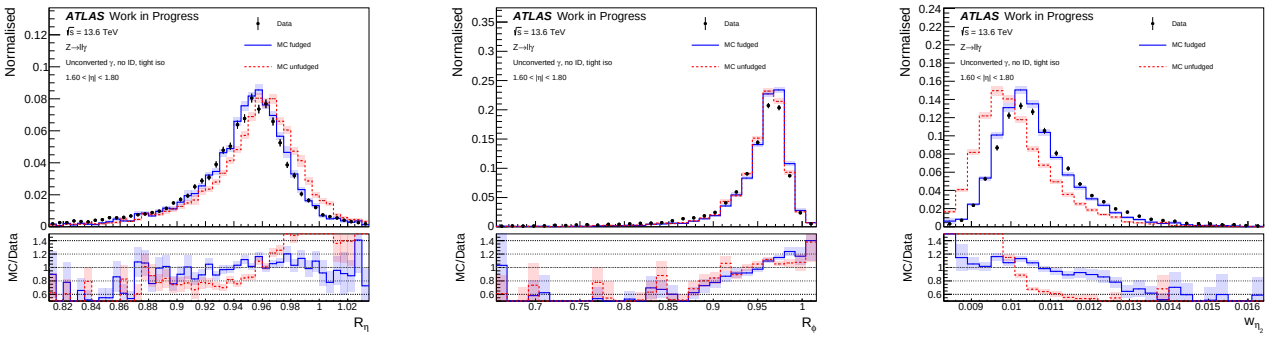
$|\eta| \in [1.52, 1.60)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 38: Fudge factor comparison, $|\eta| \in [1.52, 1.60)$.

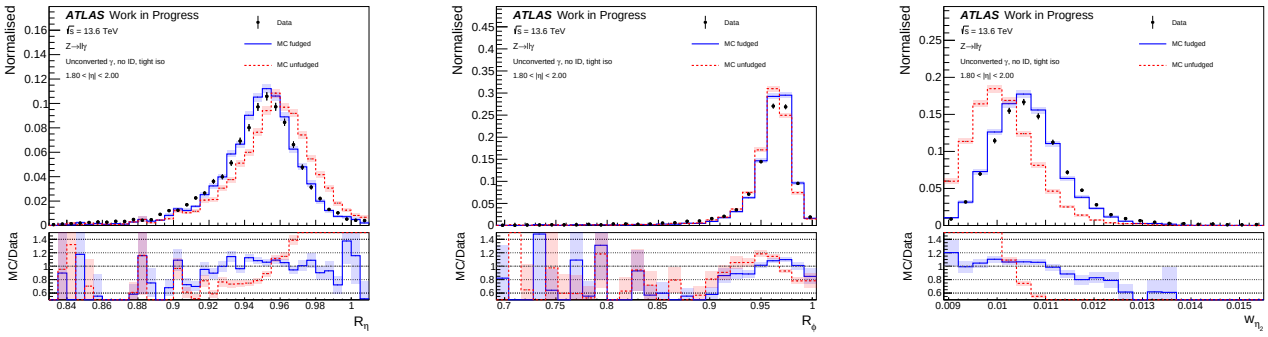
$|\eta| \in [1.60, 1.80)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 39: Fudge factor comparison, $|\eta| \in [1.60, 1.80)$.

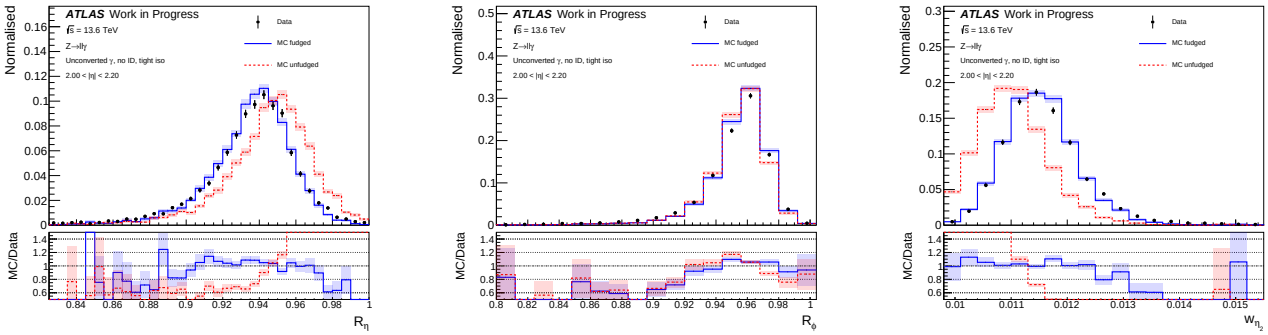
$|\eta| \in [1.80, 2.00)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 40: Fudge factor comparison, $|\eta| \in [1.80, 2.00)$.

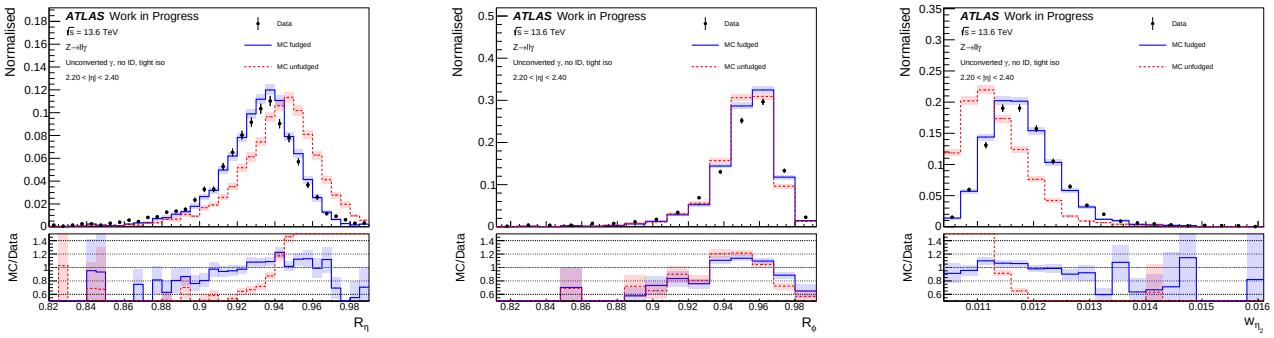
$|\eta| \in [2.00, 2.20)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 41: Fudge factor comparison, $|\eta| \in [2.00, 2.20)$.

$|\eta| \in [2.20, 2.40)$

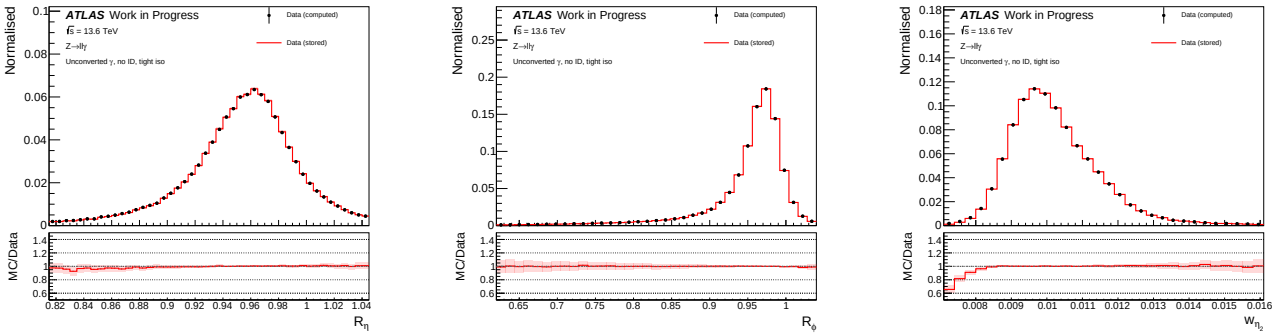



(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 42: Fudge factor comparison, $|\eta| \in [2.20, 2.40)$.

# 7 Computed vs. Stored Validation (Integrated)

Shower shapes computed from cell energies versus the branch-stored values.



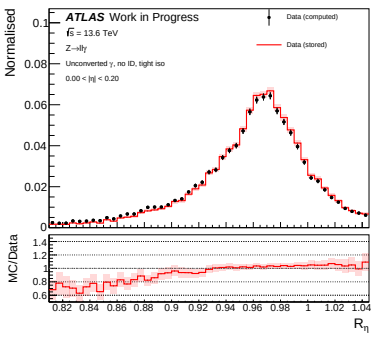

(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 43: Computed (from cells) vs stored (branch) shower shapes.

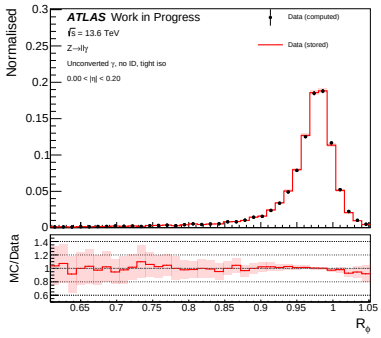
# 8 Computed vs. Stored (Per $|\eta|$ Bin)

Cell-computed vs branch-stored data per $|\eta|$ bin.

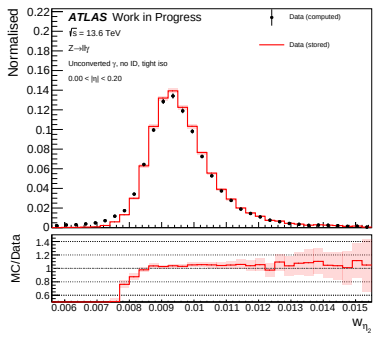
$|\eta| \in [0.00, 0.20)$




(a) $R_\eta$



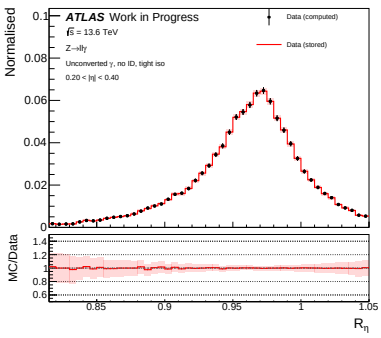

(b) $R_\phi$



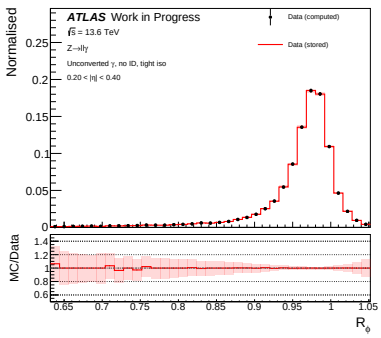

(c) $w_{\eta 2}$

Figure 44: Computed vs. stored, $|\eta| \in [0.00, 0.20)$.

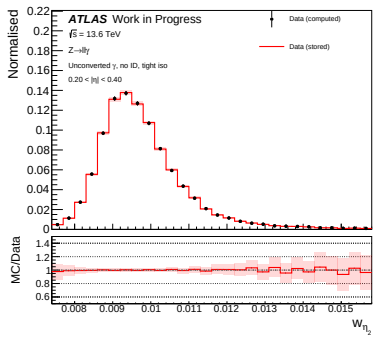
$|\eta| \in [0.20, 0.40)$




(a) $R_\eta$



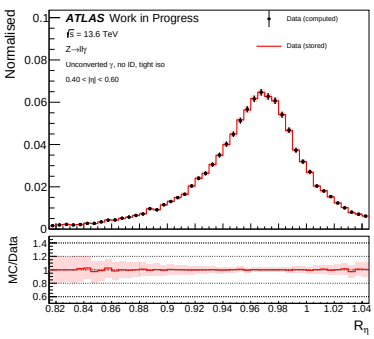

(b) $R_\phi$



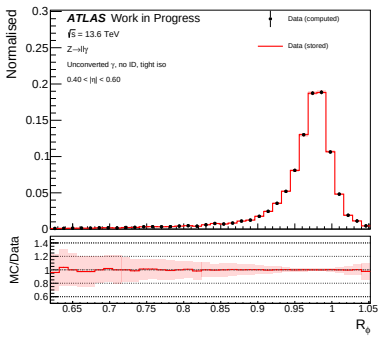

(c) $w_{\eta 2}$

Figure 45: Computed vs. stored, $|\eta| \in [0.20, 0.40)$.

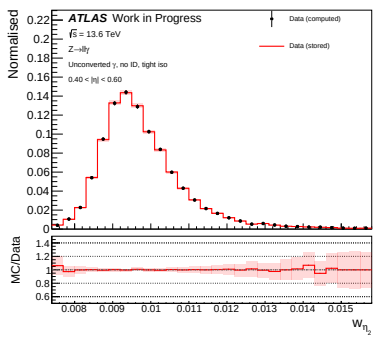
$|\eta| \in [0.40, 0.60)$




(a) $R_\eta$



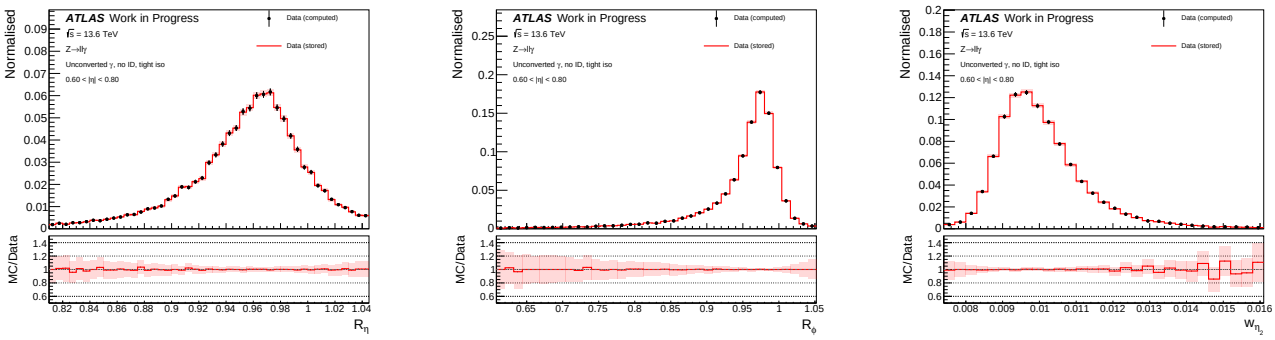

(b) $R_\phi$




(c) $w_{\eta 2}$

Figure 46: Computed vs. stored, $|\eta| \in [0.40, 0.60)$.

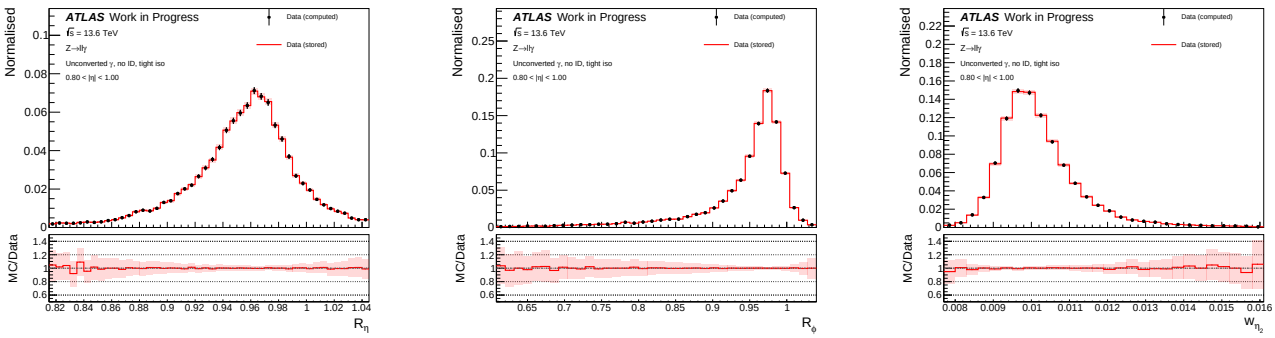
$|\eta| \in [0.60, 0.80)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 47: Computed vs. stored, $|\eta| \in [0.60, 0.80)$.

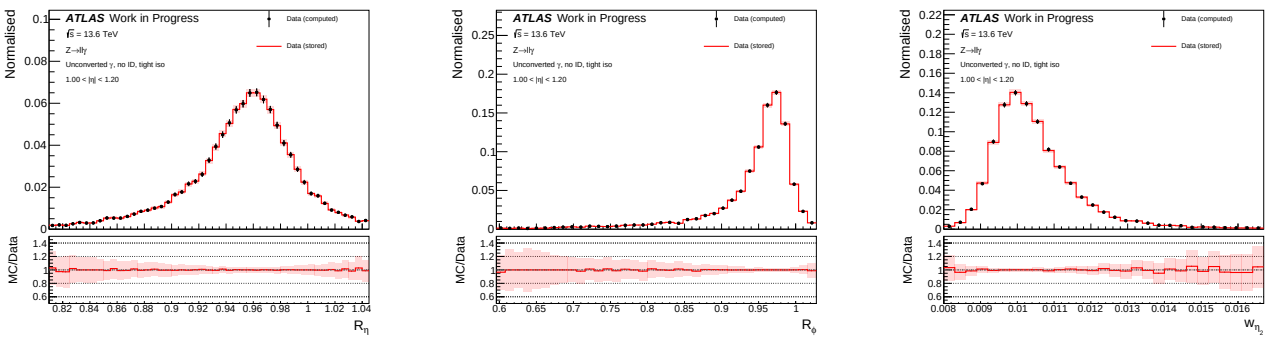
$|\eta| \in [0.80, 1.00)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 48: Computed vs. stored, $|\eta| \in [0.80, 1.00)$.

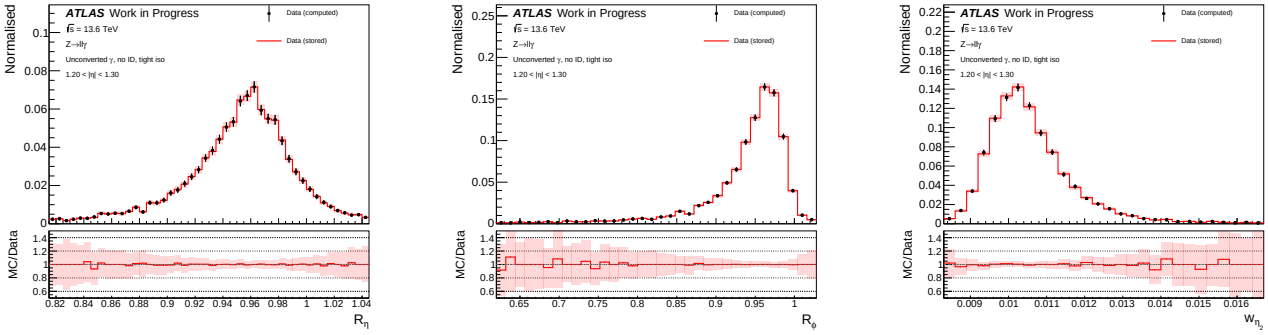
$|\eta| \in [1.00, 1.20)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 49: Computed vs. stored, $|\eta| \in [1.00, 1.20)$.

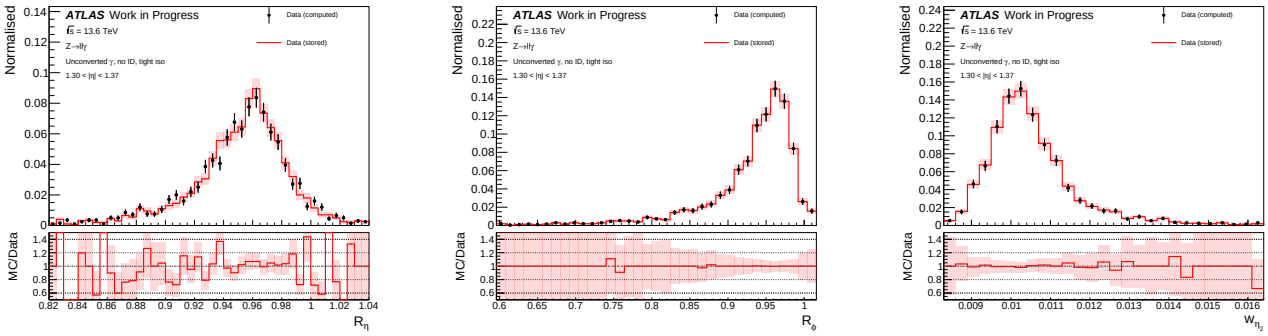
$|\eta| \in [1.20, 1.30)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 50: Computed vs. stored, $|\eta| \in [1.20, 1.30)$.

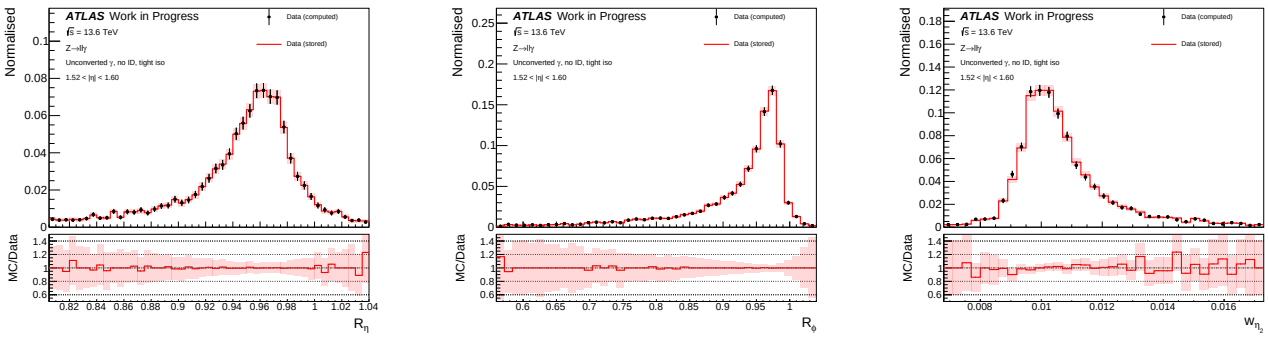
$|\eta| \in [1.30, 1.37)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 51: Computed vs. stored, $|\eta| \in [1.30, 1.37)$.

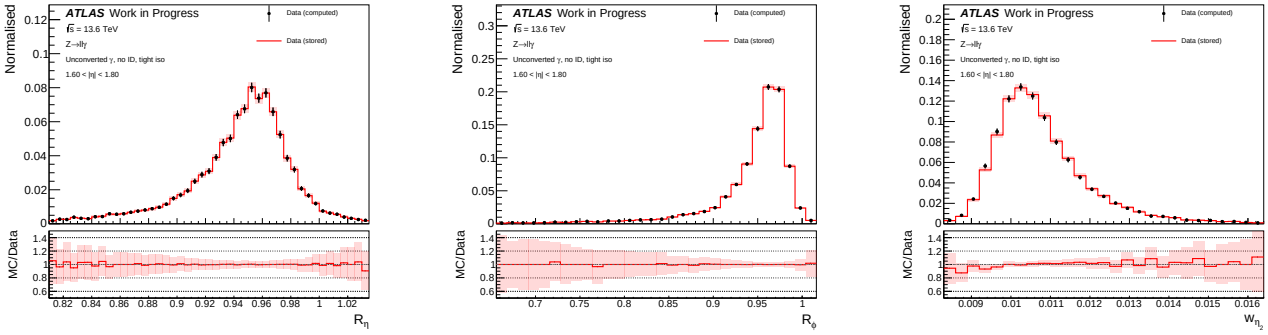
$|\eta| \in [1.52, 1.60)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 52: Computed vs. stored, $|\eta| \in [1.52, 1.60)$.

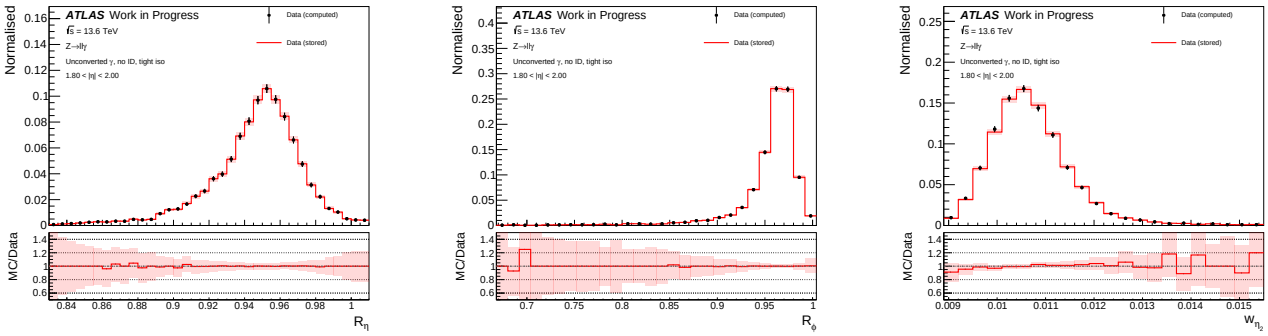
$|\eta| \in [1.60, 1.80)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 53: Computed vs. stored, $|\eta| \in [1.60, 1.80)$.

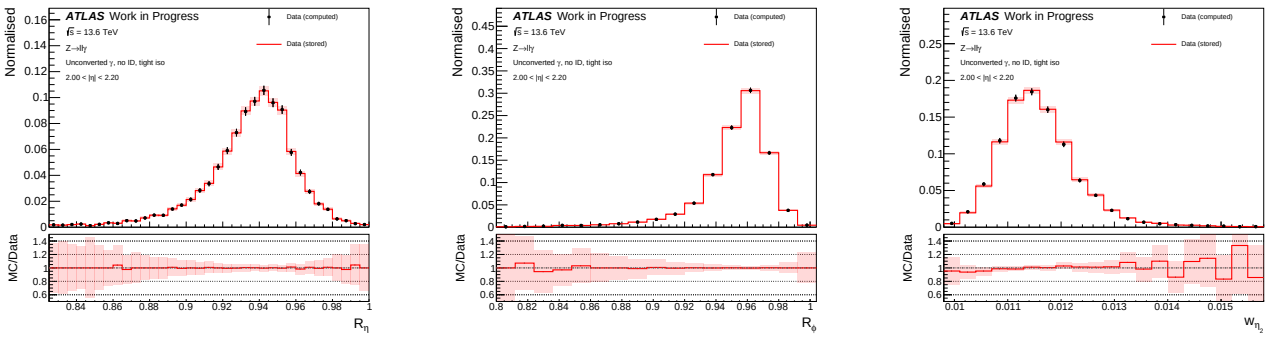
$|\eta| \in [1.80, 2.00)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 54: Computed vs. stored, $|\eta| \in [1.80, 2.00)$.

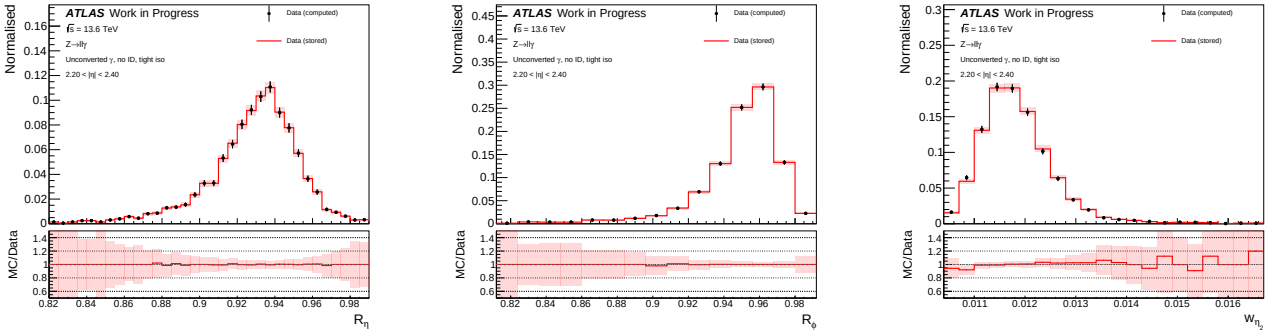
$|\eta| \in [2.00, 2.20)$




(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 55: Computed vs. stored, $|\eta| \in [2.00, 2.20)$.

$|\eta| \in [2.20, 2.40)$



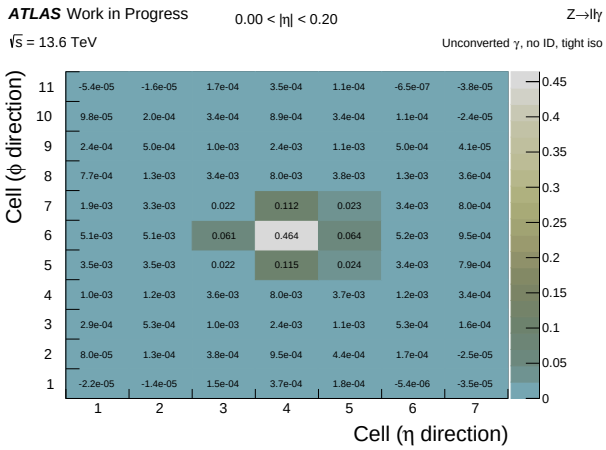

(a) $R_\eta$ (b) $R_\phi$ (c) $w_{\eta 2}$

Figure 56: Computed vs. stored, $|\eta| \in [2.20, 2.40)$.

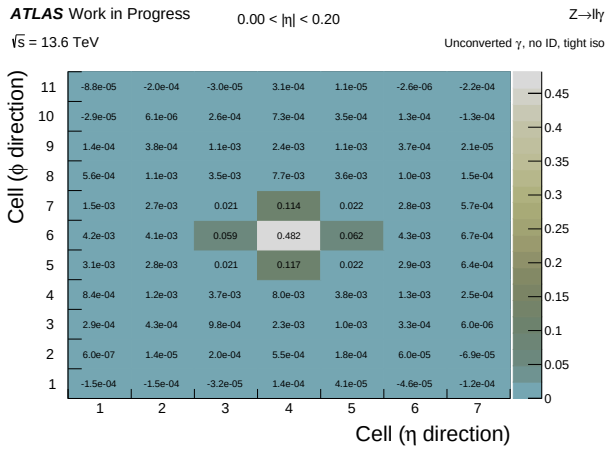
# 9 Cell Energy Fraction Profiles

Mean cell energy fraction $\langle f_k \rangle$ maps in the $7 \times 11$ $(\eta \times \phi)$ grid. Each page shows one $|\eta|$ bin.

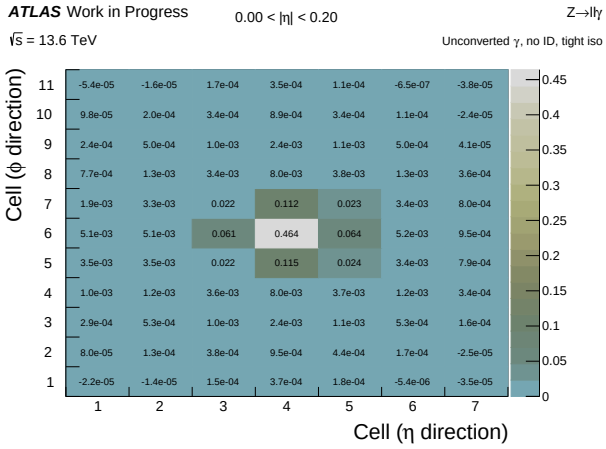
$|\eta| \in [0.00, 0.20)$



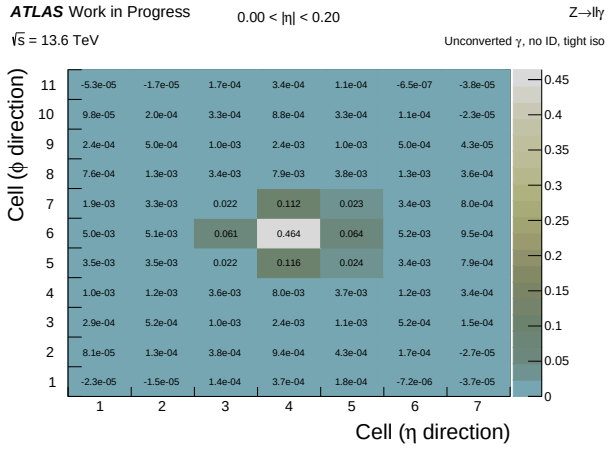

(a) Data




(b) MC (uncorrected)



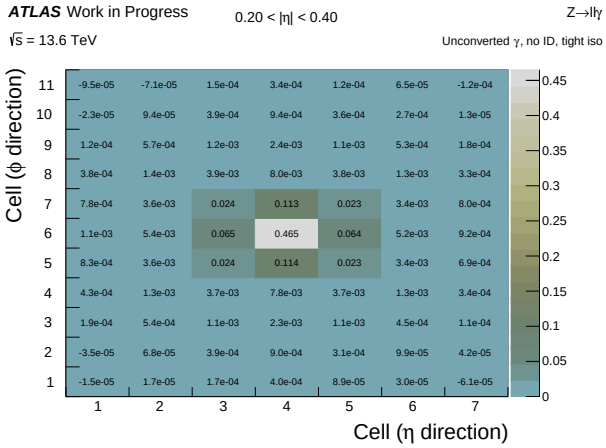

(c) MC after M1



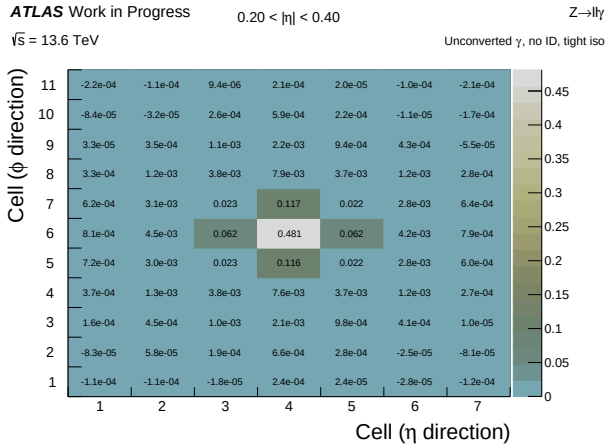

(d) MC after M2

Figure 57: Cell energy fraction maps, $|\eta| \in [0.00, 0.20)$.

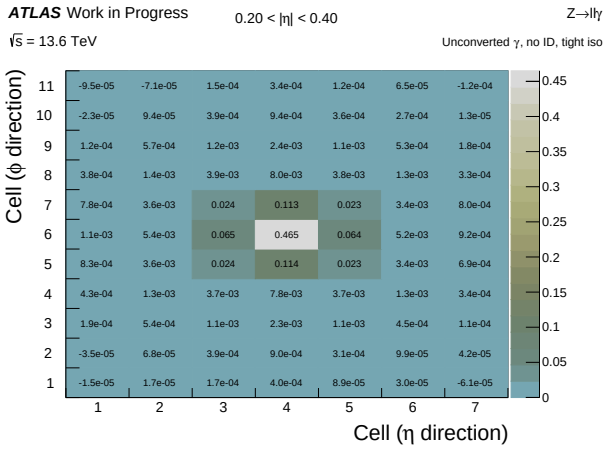
$|\eta| \in [0.20, 0.40)$



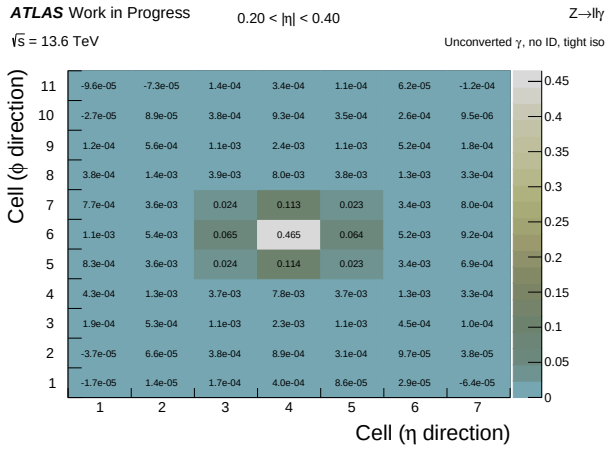

(a) Data




(b) MC (uncorrected)



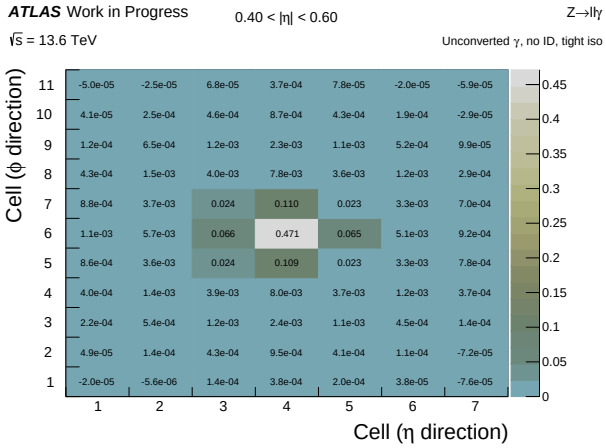

(c) MC after M1



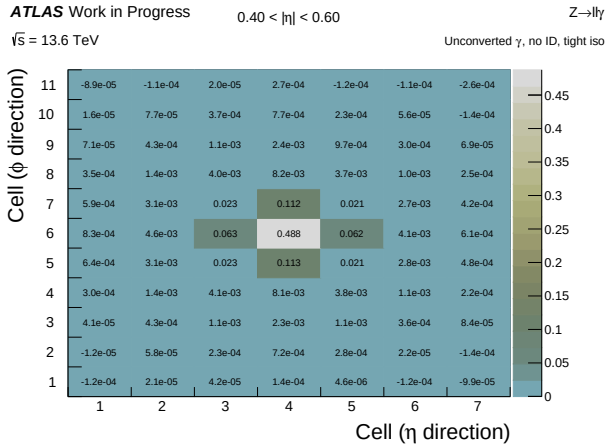

(d) MC after M2

Figure 58: Cell energy fraction maps, $|\eta| \in [0.20, 0.40)$.

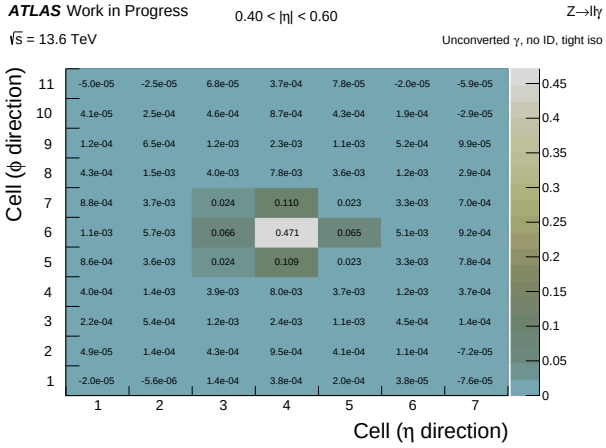
$|\eta| \in [0.40, 0.60)$



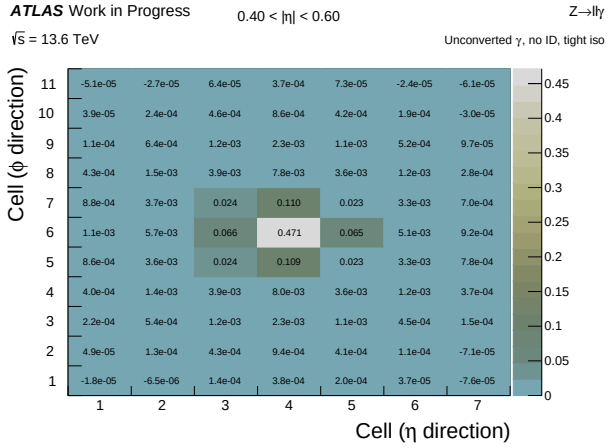

(a) Data




(b) MC (uncorrected)



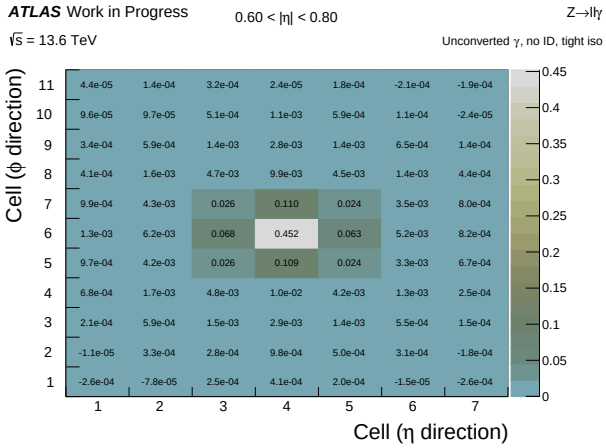

(c) MC after M1



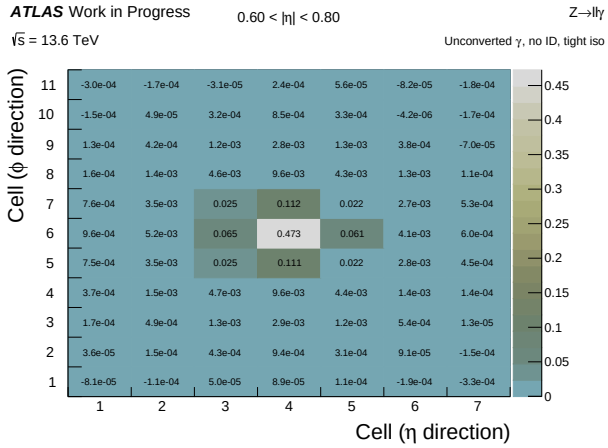

(d) MC after M2

Figure 59: Cell energy fraction maps, $|\eta| \in [0.40, 0.60)$.

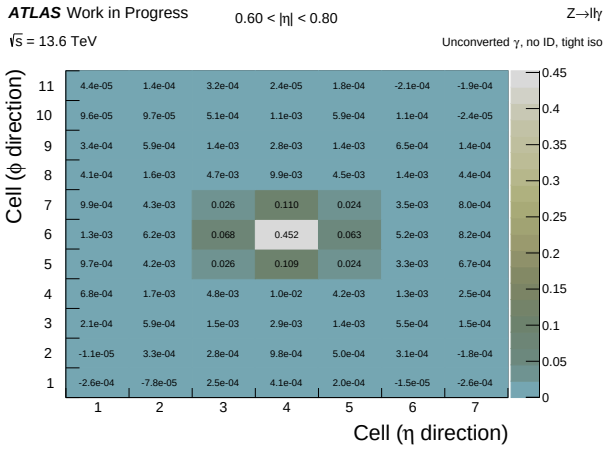
$|\eta| \in [0.60, 0.80)$



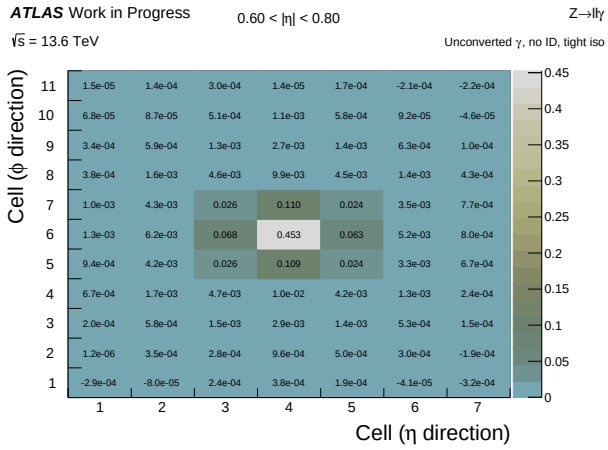

(a) Data




(b) MC (uncorrected)



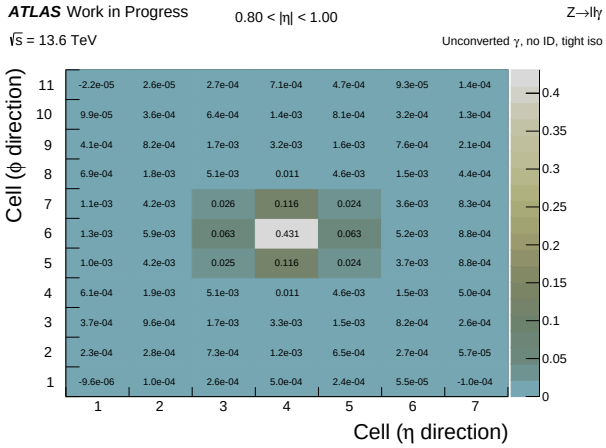

(c) MC after M1



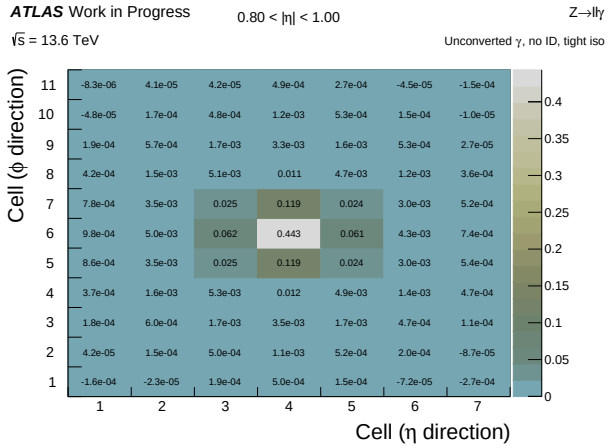

(d) MC after M2

Figure 60: Cell energy fraction maps, $|\eta| \in [0.60, 0.80)$.

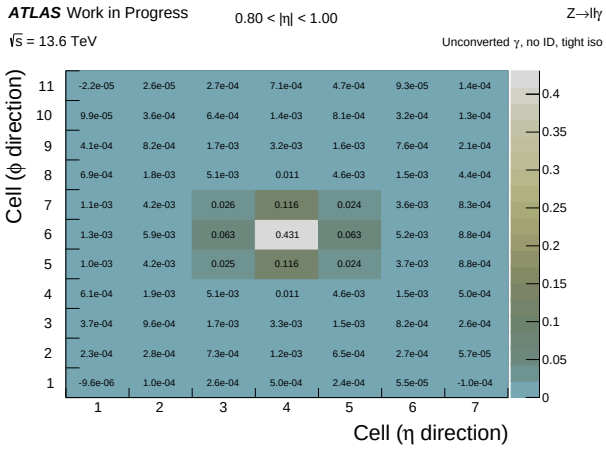
$|\eta| \in [0.80, 1.00)$



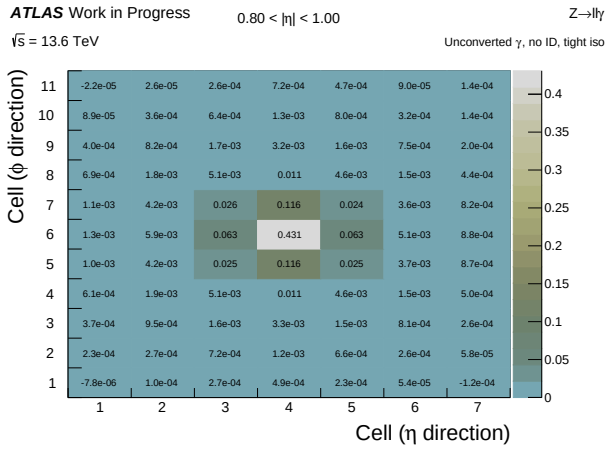

(a) Data




(b) MC (uncorrected)



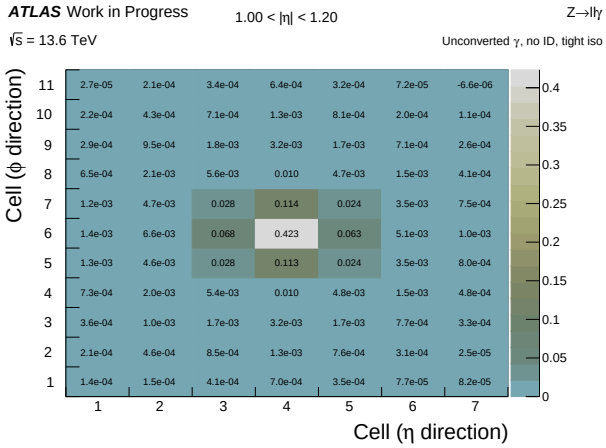

(c) MC after M1



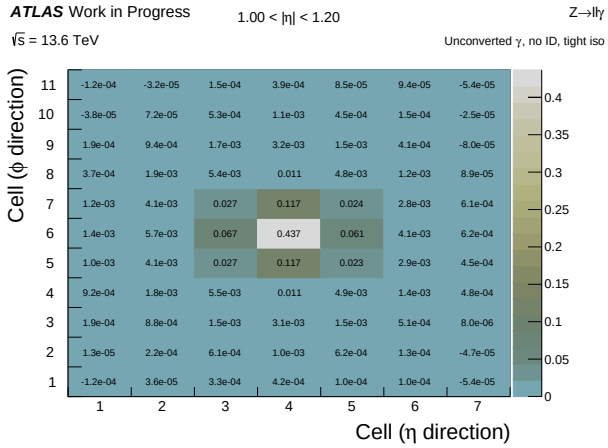

(d) MC after M2

Figure 61: Cell energy fraction maps, $|\eta| \in [0.80, 1.00)$.

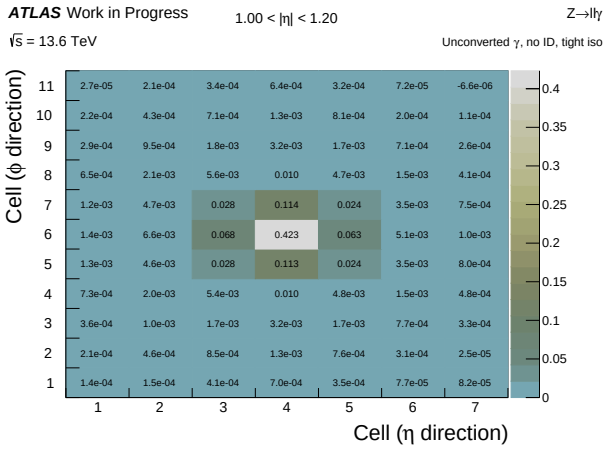
$|\eta| \in [1.00, 1.20)$



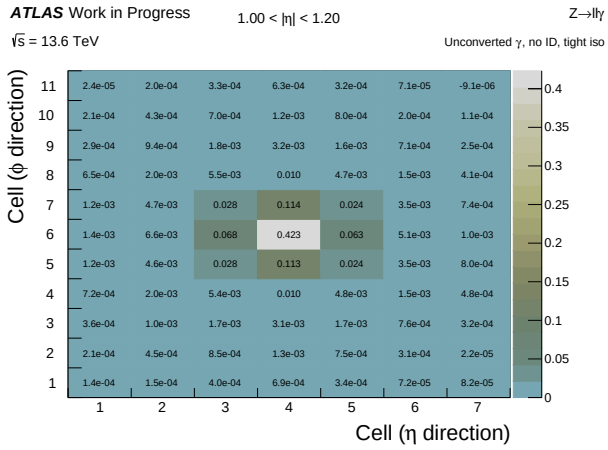

(a) Data




(b) MC (uncorrected)



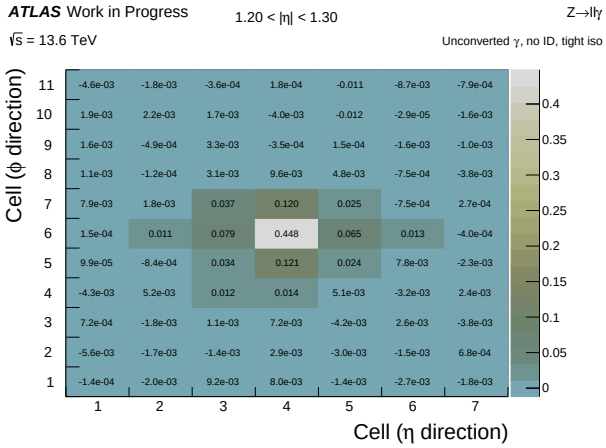

(c) MC after M1



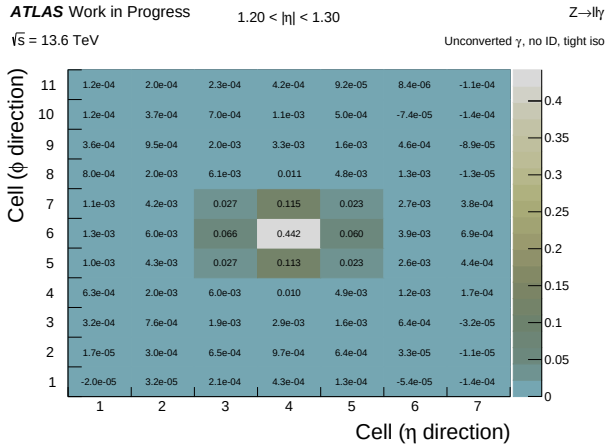

(d) MC after M2

Figure 62: Cell energy fraction maps, $|\eta| \in [1.00, 1.20)$.

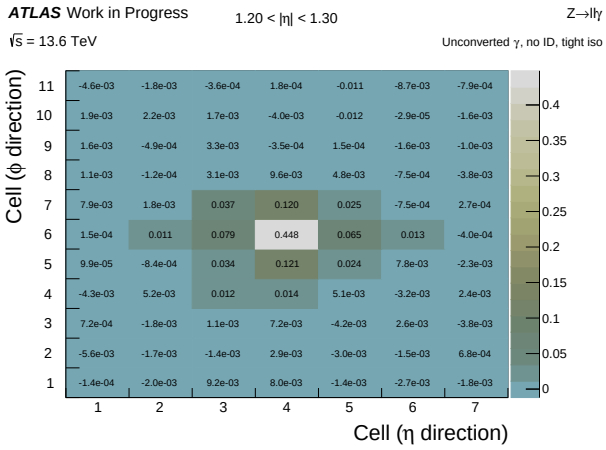
$|\eta| \in [1.20, 1.30)$



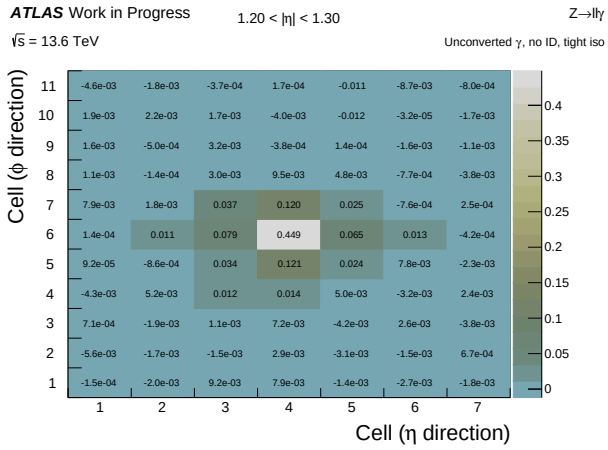

(a) Data




(b) MC (uncorrected)



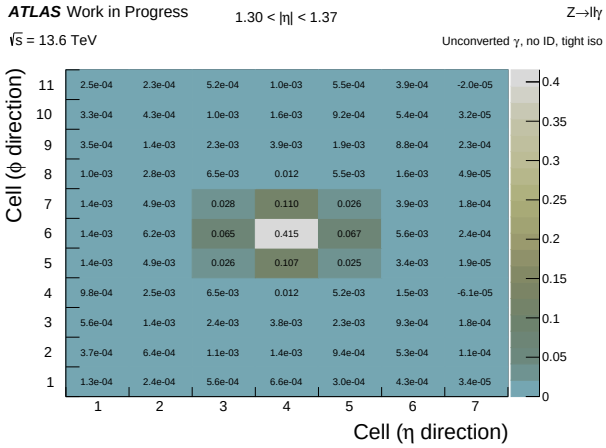

(c) MC after M1



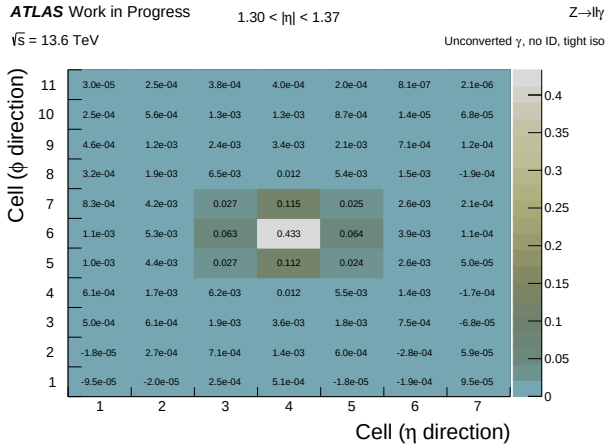

(d) MC after M2

Figure 63: Cell energy fraction maps, $|\eta| \in [1.20, 1.30)$.

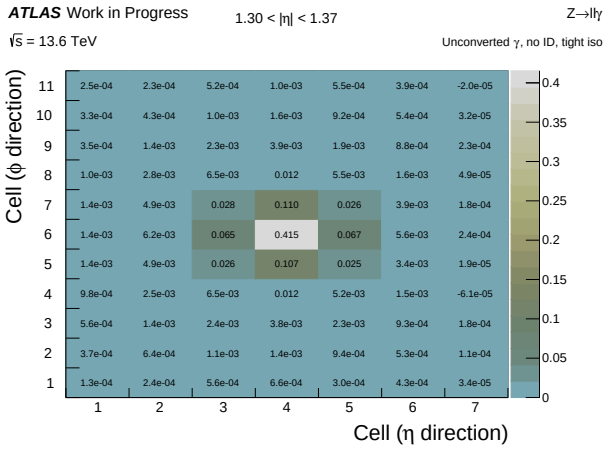
$|\eta| \in [1.30, 1.37)$



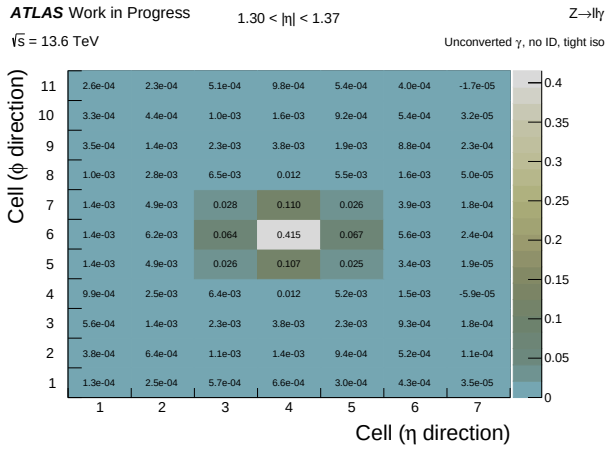

(a) Data




(b) MC (uncorrected)



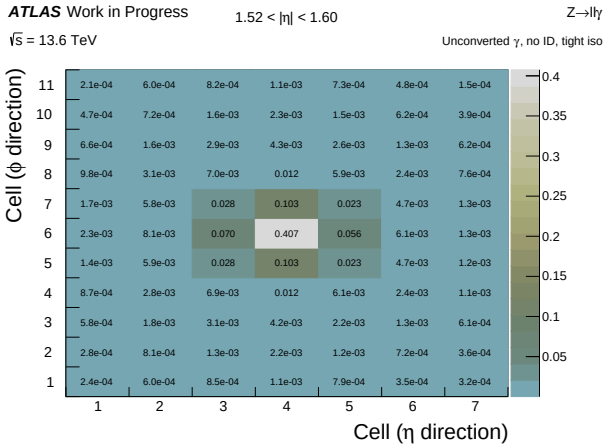

(c) MC after M1



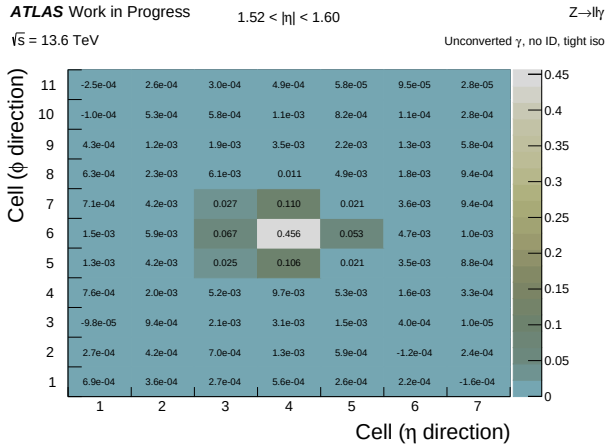

(d) MC after M2

Figure 64: Cell energy fraction maps, $|\eta| \in [1.30, 1.37)$.

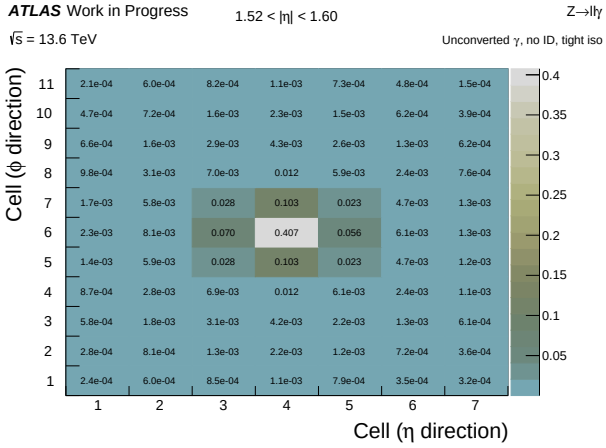
$|\eta| \in [1.52, 1.60)$



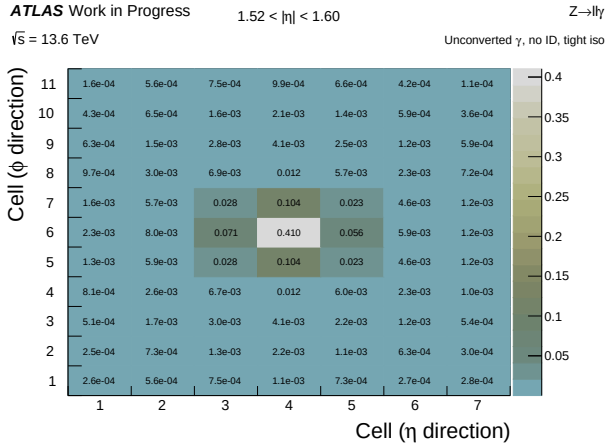

(a) Data




(b) MC (uncorrected)



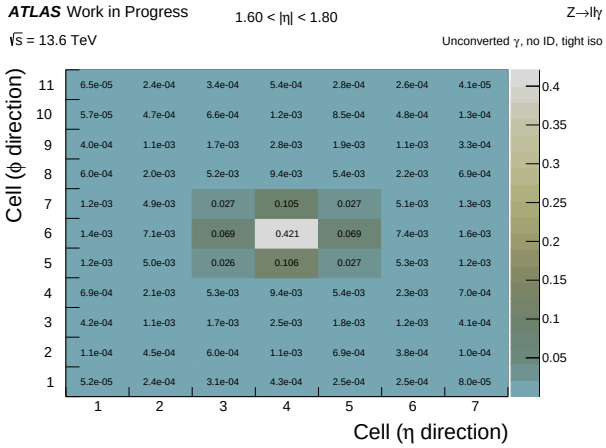

(c) MC after M1



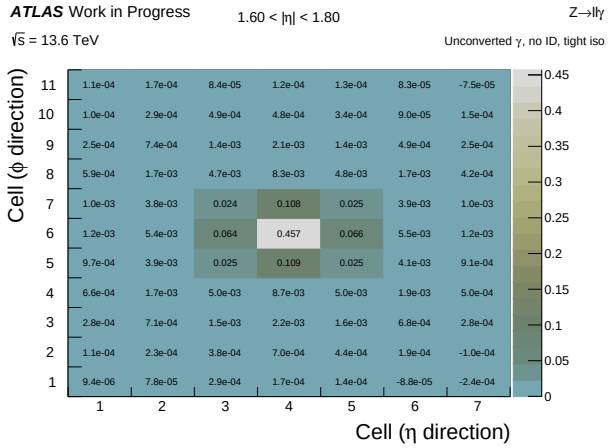

(d) MC after M2

Figure 65: Cell energy fraction maps, $|\eta| \in [1.52, 1.60)$.

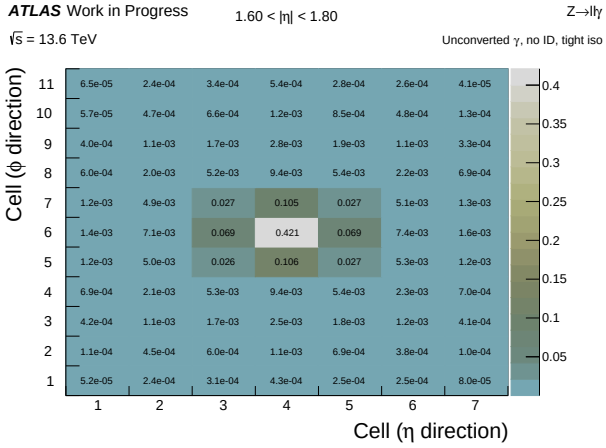
$|\eta| \in [1.60, 1.80)$



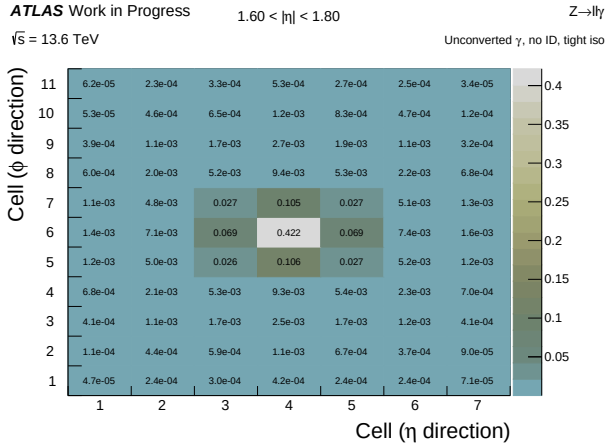

(a) Data




(b) MC (uncorrected)



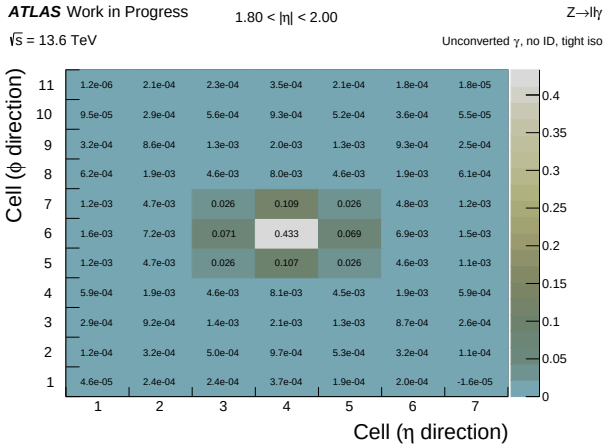

(c) MC after M1



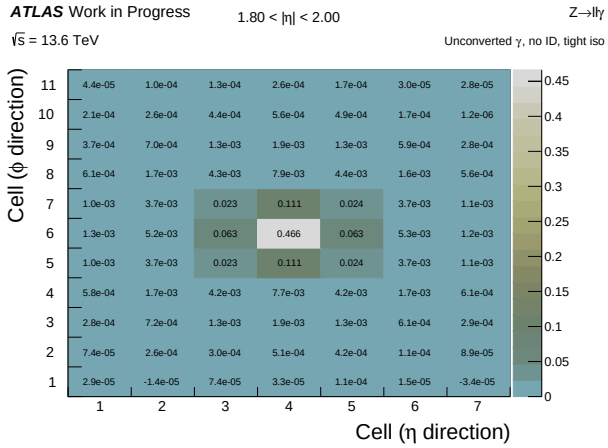

(d) MC after M2

Figure 66: Cell energy fraction maps, $|\eta| \in [1.60, 1.80)$.

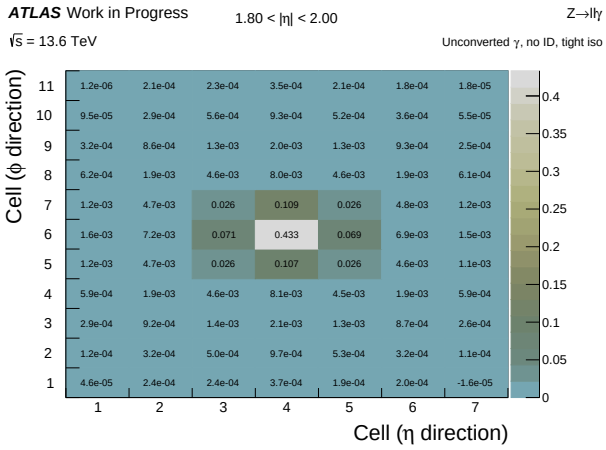
$|\eta| \in [1.80, 2.00)$



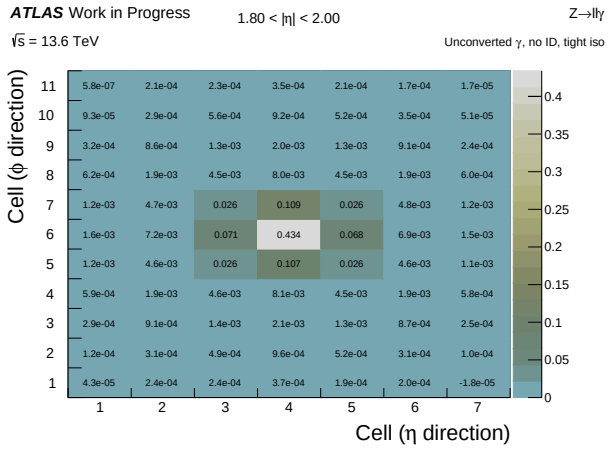

(a) Data




(b) MC (uncorrected)



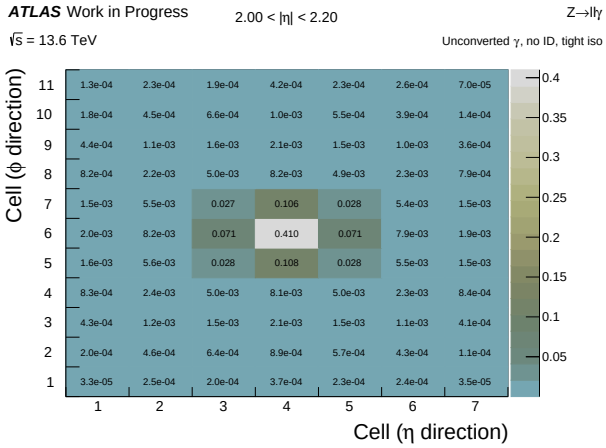

(c) MC after M1



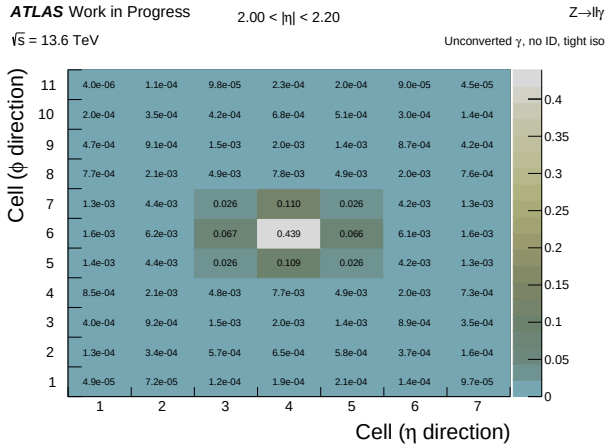

(d) MC after M2

Figure 67: Cell energy fraction maps, $|\eta| \in [1.80, 2.00)$.

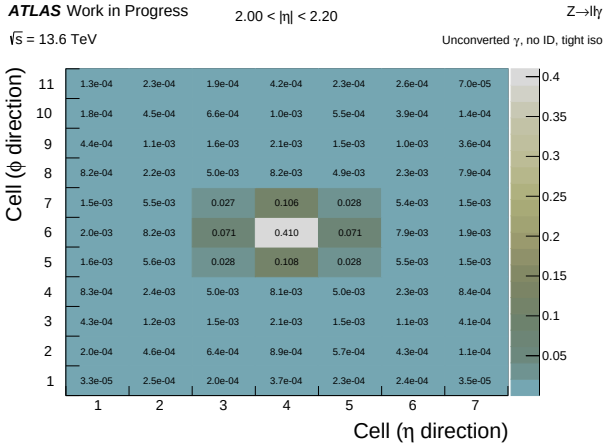
$|\eta| \in [2.00, 2.20)$



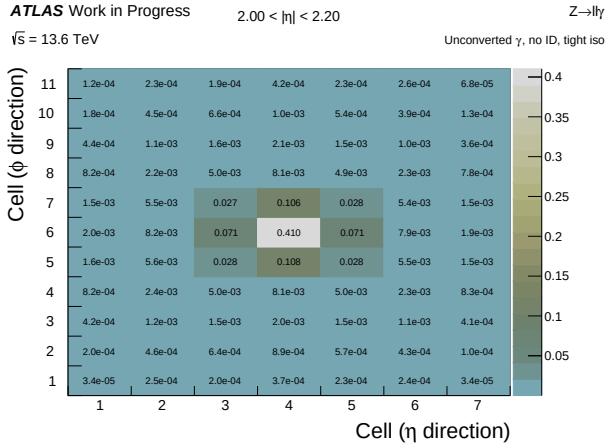

(a) Data




(b) MC (uncorrected)



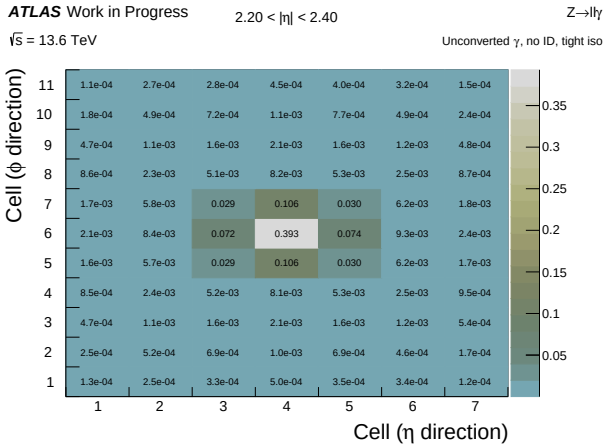

(c) MC after M1



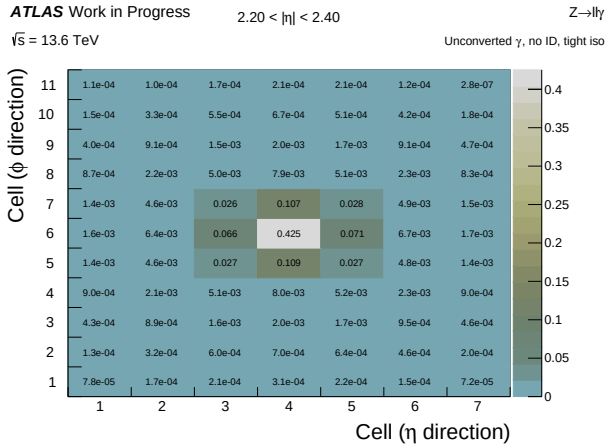

(d) MC after M2

Figure 68: Cell energy fraction maps, $|\eta| \in [2.00, 2.20)$.

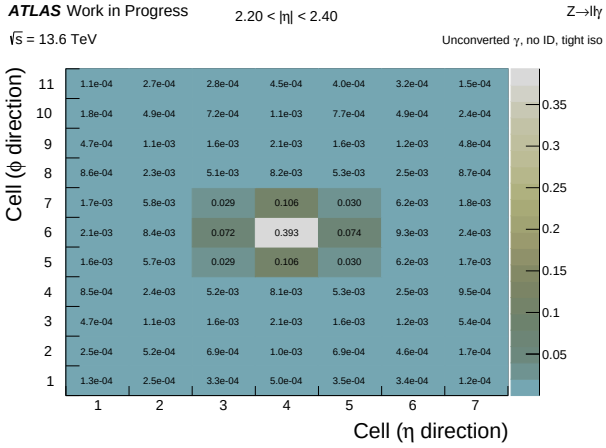
$|\eta| \in [2.20, 2.40)$



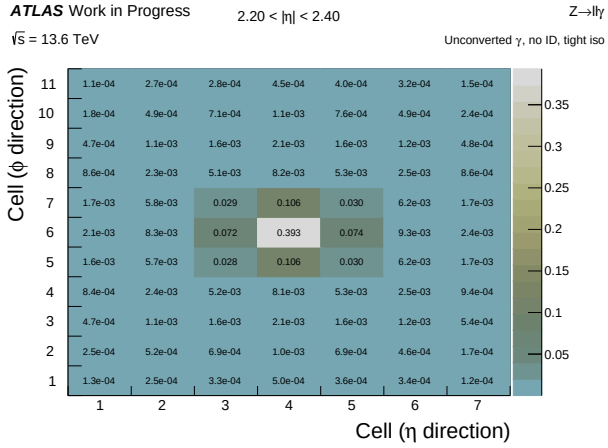

(a) Data




(b) MC (uncorrected)




(c) MC after M1



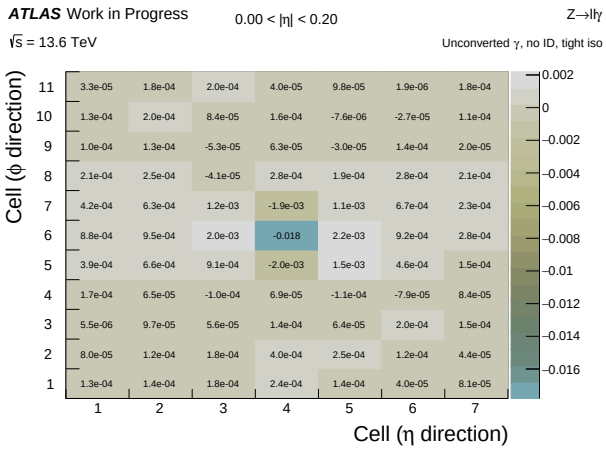

(d) MC after M2

Figure 69: Cell energy fraction maps, $|\eta| \in [2.20, 2.40)$.

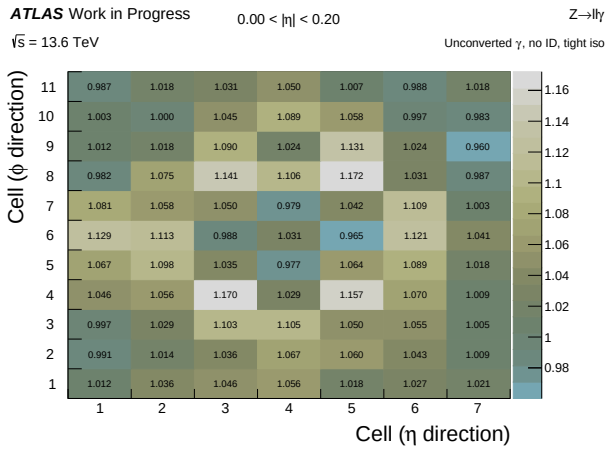
# 10 Cell-Level Corrections

M1 shift ($\Delta_k$) and M2 stretch ($s_k$) maps per $|\eta|$ bin.

$|\eta| \in [0.00, 0.20)$



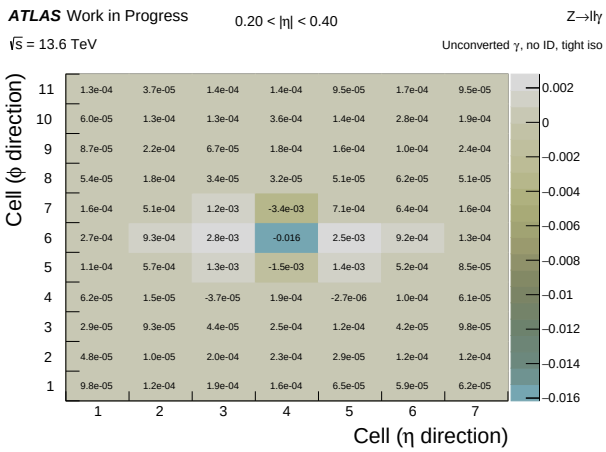

(a) M2 shift ($a_k$)



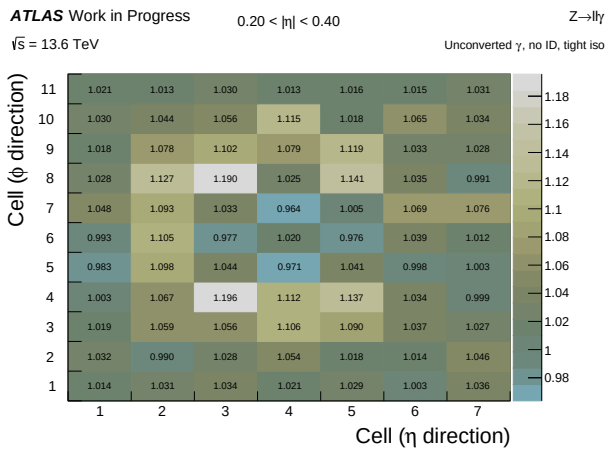

(b) M2 stretch ($s_k$)

Figure 70: Correction maps, $|\eta| \in [0.00, 0.20)$.

$|\eta| \in [0.20, 0.40)$



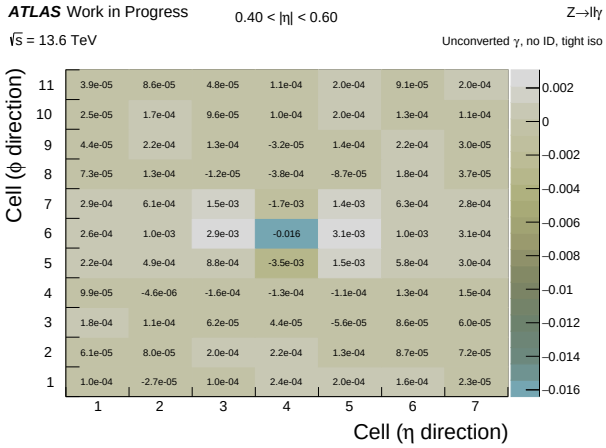

(a) M2 shift ($a_k$)



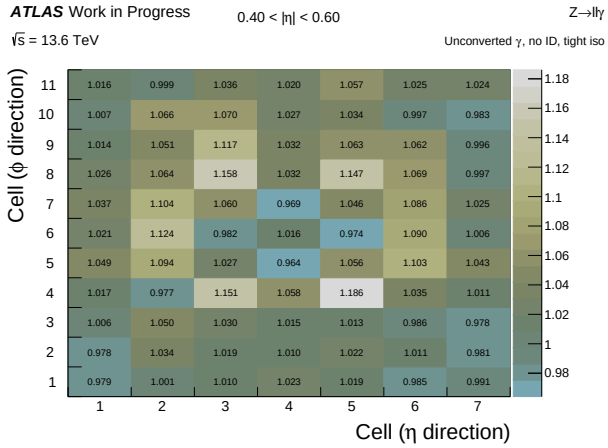

(b) M2 stretch ($s_k$)

Figure 71: Correction maps, $|\eta| \in [0.20, 0.40)$.

$|\eta| \in [0.40, 0.60)$



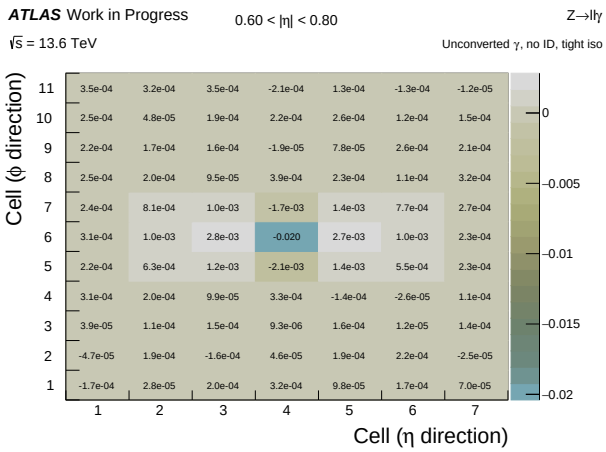

(a) M2 shift ($a_k$)



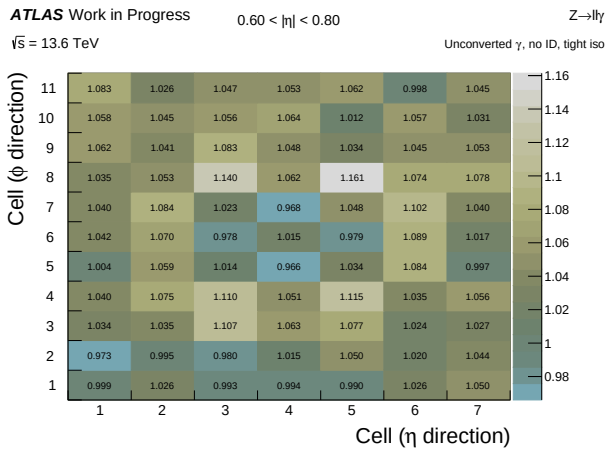

(b) M2 stretch ($s_k$)

Figure 72: Correction maps, $|\eta| \in [0.40, 0.60)$.

$|\eta| \in [0.60, 0.80)$



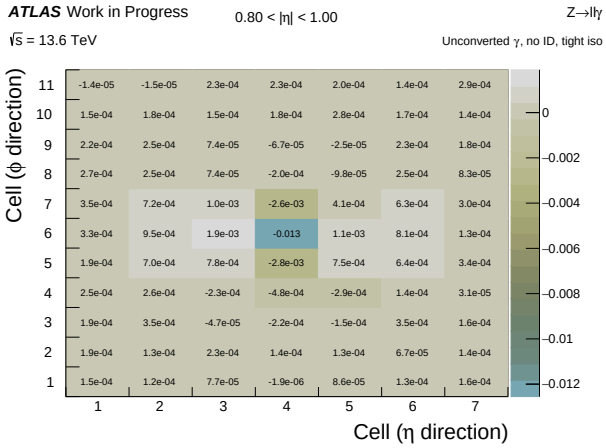

(a) M2 shift ($a_k$)



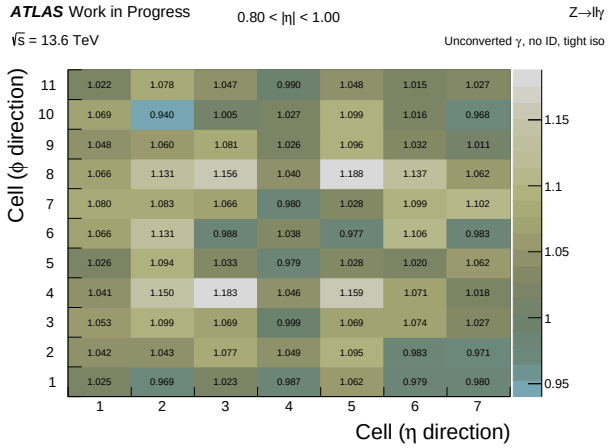

(b) M2 stretch ($s_k$)

Figure 73: Correction maps, $|\eta| \in [0.60, 0.80)$.

$|\eta| \in [0.80, 1.00)$



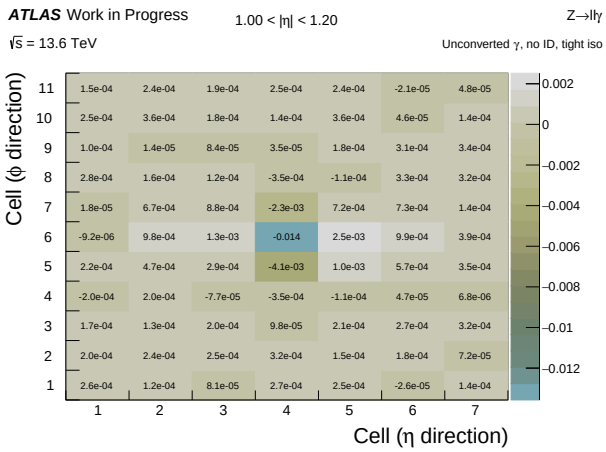

(a) M2 shift ($a_k$)



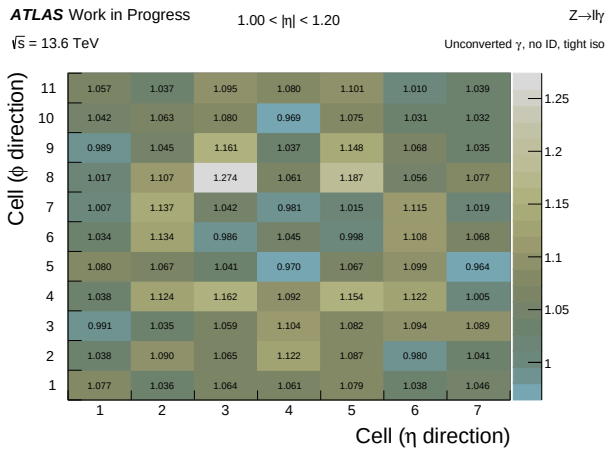

(b) M2 stretch ($s_k$)

Figure 74: Correction maps, $|\eta| \in [0.80, 1.00)$.

$|\eta| \in [1.00, 1.20)$



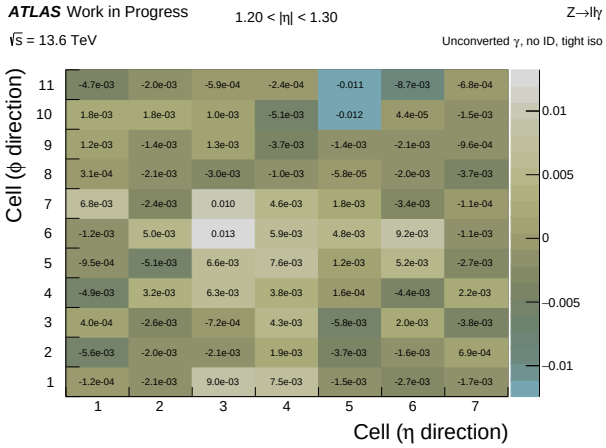

(a) M2 shift ($a_k$)



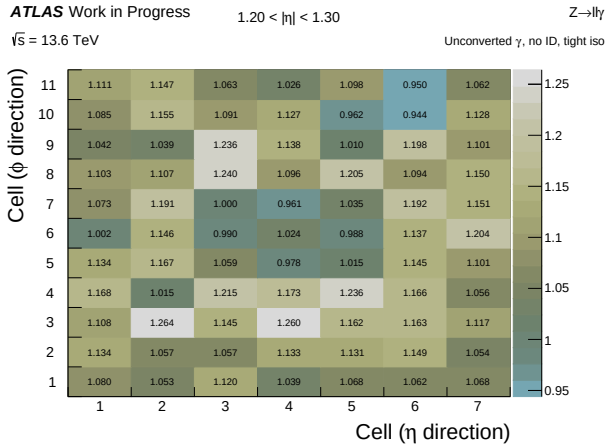

(b) M2 stretch ($s_k$)

Figure 75: Correction maps, $|\eta| \in [1.00, 1.20)$.

$|\eta| \in [1.20, 1.30)$



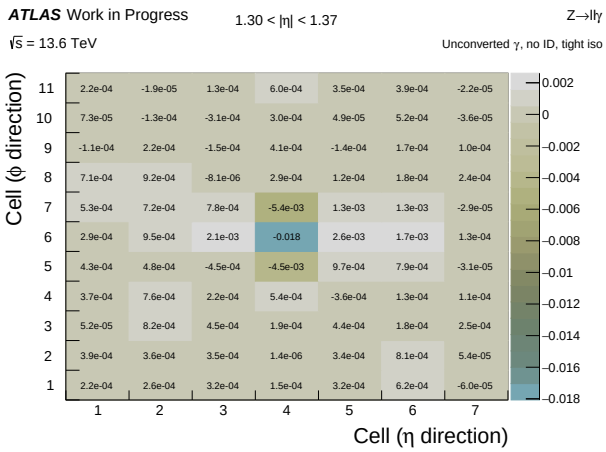

(a) M2 shift ($a_k$)



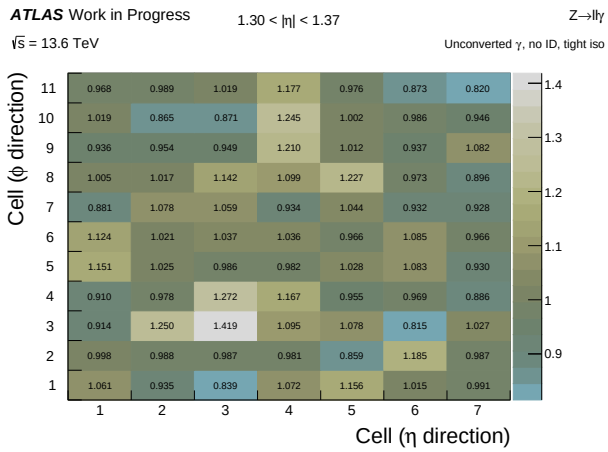

(b) M2 stretch ($s_k$)

Figure 76: Correction maps, $|\eta| \in [1.20, 1.30)$.

$|\eta| \in [1.30, 1.37)$



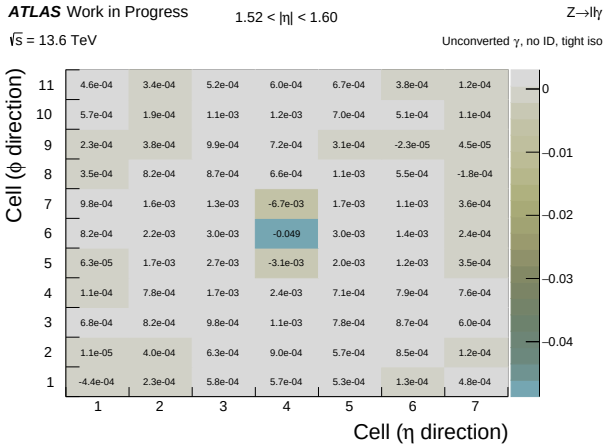

(a) M2 shift ($a_k$)



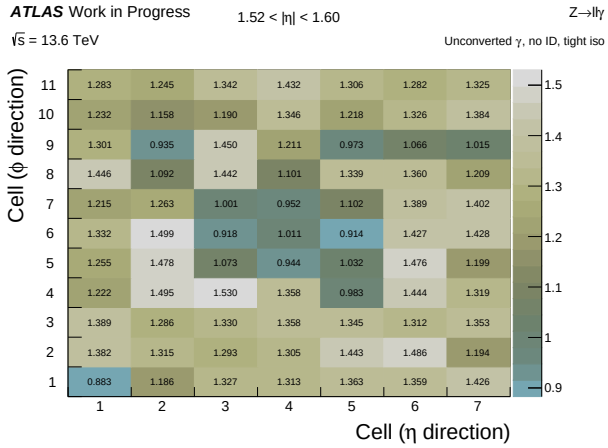

(b) M2 stretch ($s_k$)

Figure 77: Correction maps, $|\eta| \in [1.30, 1.37)$.

$|\eta| \in [1.52, 1.60)$



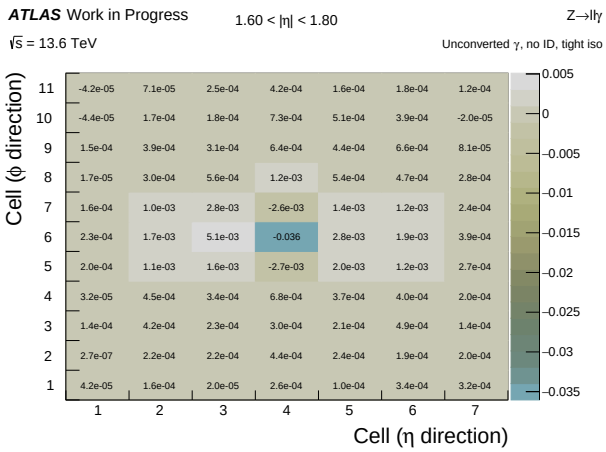

(a) M2 shift ($a_k$)



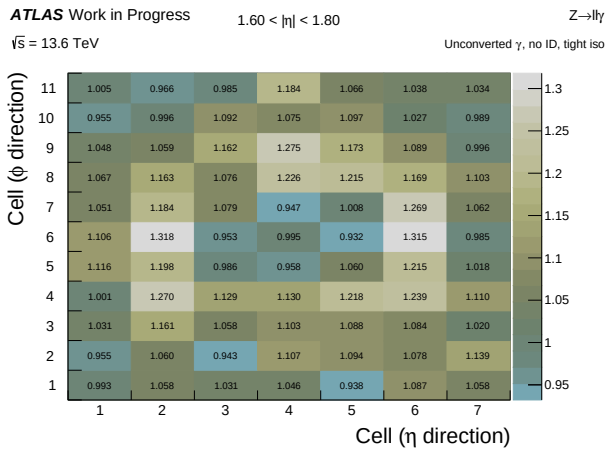

(b) M2 stretch ($s_k$)

Figure 78: Correction maps, $|\eta| \in [1.52, 1.60)$.

$|\eta| \in [1.60, 1.80)$



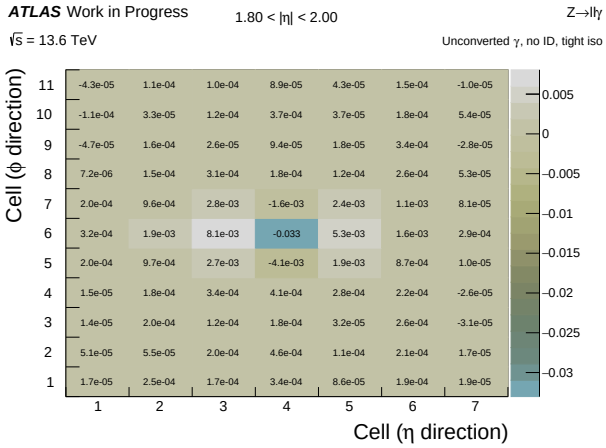

(a) M2 shift ($a_k$)



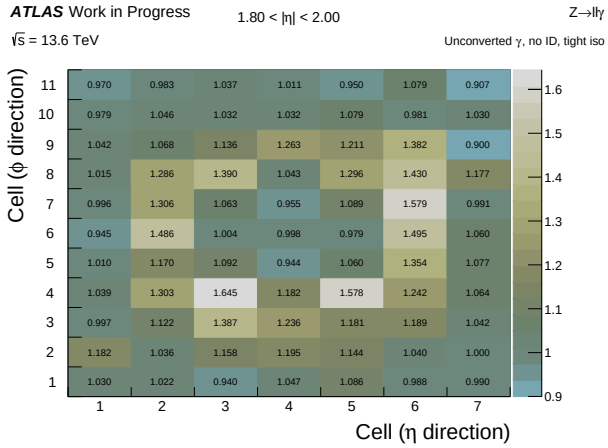

(b) M2 stretch ($s_k$)

Figure 79: Correction maps, $|\eta| \in [1.60, 1.80)$.

$|\eta| \in [1.80, 2.00)$



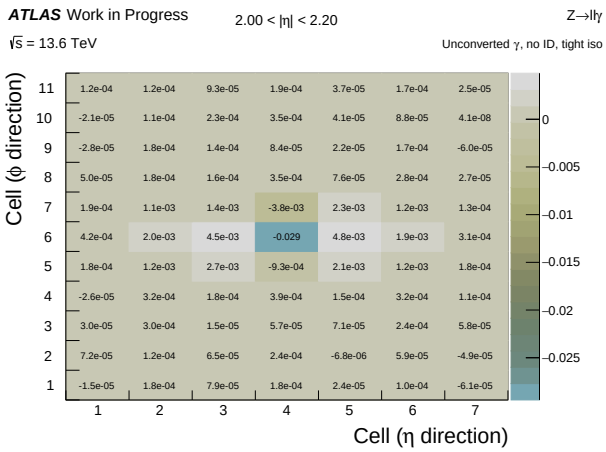

(a) M2 shift ($a_k$)



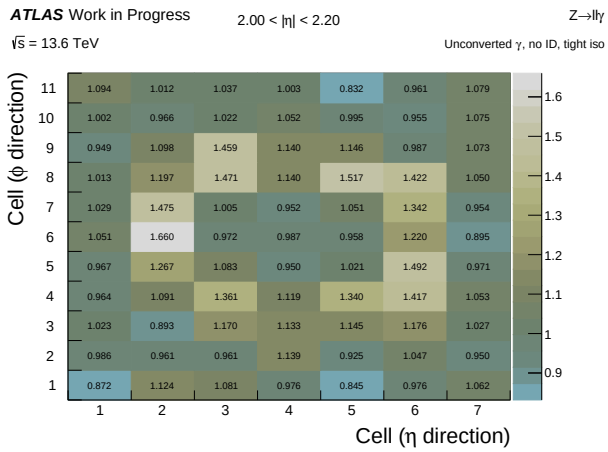

(b) M2 stretch ($s_k$)

Figure 80: Correction maps, $|\eta| \in [1.80, 2.00)$.

$|\eta| \in [2.00, 2.20)$



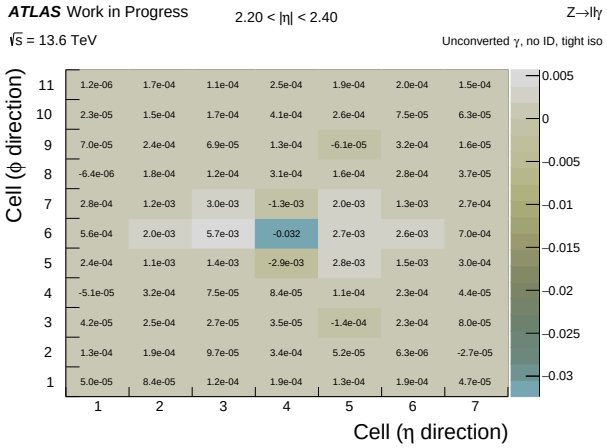

(a) M2 shift ($a_k$)



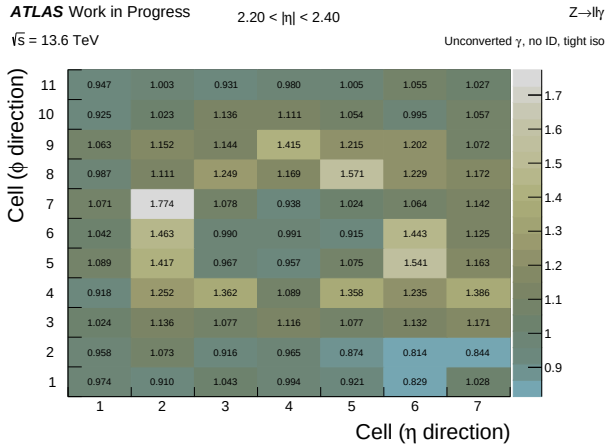

(b) M2 stretch ($s_k$)

Figure 81: Correction maps, $|\eta| \in [2.00, 2.20)$.

$|\eta| \in [2.20, 2.40)$




(a) M2 shift ($a_k$)




(b) M2 stretch ($s_k$)

Figure 82: Correction maps, $|\eta| \in [2.20, 2.40)$.